NeedleFinder 2026: Sub-Millimeter Catheter Reconstruction on MRI for Interstitial Gynecologic Brachytherapy in Seconds

Guillaume Pernelle, Paolo Zaffino, Lauren Barber, Alireza Mehrtash, Christiane Yeung, Tina Kapur, and Collaborators

October 2026 (Target Journal: Physics in Medicine & Biology*)*

ABSTRACT

Objective. Interstitial gynecologic brachytherapy planned on MRI alone requires locating 15 to 30 thin plastic catheters on 3T T2-weighted images. Tracing them by hand takes tens of minutes while the patient is under anesthesia. Automatic methods based on voxel segmentation break catheters into fragments and mistake blood vessels for catheters. We set out to make catheter reconstruction automatic, accurate to below a millimeter, and fast enough to run during the procedure.

Approach. NeedleFinder 2026 works in three stages. Stage 0 computes an analytical 3D circular dark-core symmetry prior (S_0) and a 100% reference-free, tilt-safe axial field-of-view bound (`k_lim_auto`) that preserves 100.0% of in-FOV catheter depth. Stage 1 is a native-resolution 3D residual U-Net (`NeedleUNet3D`) self-supervised pre-trained on 51 strictly external pelvic T2W MRI volumes from 50 distinct subjects (combining 3D masked autoencoding with synthetic 3D susceptibility-tube pretext learning against real anatomical hard negatives) and trained with multi-rater supervision (`CY+HT` plus consensus-fused and soft-weighted `LB/AMe` traces on outer-training folds), 5-fold inner ensembling, and 4-view 3D test-time augmentation. Stage 2 traces candidate trajectories with native bidirectional Kalman tracking, a 4-layer triplet message-passing network, and collinear split-fragment stitching (`102` stitched full-length trajectories), scores the whole implant bundle with a graph transformer pre-trained on 300 simulated bundles (`SynthBrachy3T`), and refines each distal tip with a dedicated 91-step (`0.5 mm` sub-slice Z-interpolated) 22-channel 3-block dilated 1D ResNet + 4-head self-attention axial localizer (`Dedicated1DTipLocalizer`) equipped with strictly nested split-conformal 3D tip uncertainty intervals. We evaluated the pipeline on 54 clinical 3T MRI scans (1,063 catheters) under a strictly nested 5×5 patient-level cross-validation protocol (25 inner models, zero outer-test leakage), against three published methods, across all 40 dual-traced scans via Grubbs' three-cornered-hat and two-one-sided tests (TOST) for non-inferiority, in AAPM TG-43U1 HDR ^{192}Ir dosimetric simulations, on 48 public needle-free pelvic MRI scans, and zero-shot across prostate MRI (`Case 100`), secondary MRI sequences (`3D wbSSFP`, `2D AX TSE T2`), 23 serial 4D insertion scans (`Cases 68–71`), and 40 paired post-implant pelvic CT scans.

Main results. At the pre-specified threshold $\tau = 50\%$, NeedleFinder 2026 detected 91.4% of all 1,063 catheters (`972/1,063`) with 86.9% precision (`972/1,118`) and 89.1% F1 (`89.1%` with inner-OOF cross-fit; `89.4%` F1 at $\tau = 55\%$; `93.6%` recall at $\tau = 35\%$), outperforming our 2019-style 3D U-Net (`83.1%` F1, $p = 1.1 \times 10^{-5}$), a native-resolution clDice ResUNet (`38.7%` F1, $p = 5.2 \times 10^{-10}$), and a 2026-style structure-aware tracker (`54.2%` F1, $p = 3.0 \times 10^{-10}$). On the 759 long interstitial shafts, it achieved 91.3% recall (`92.5%` inner-OOF cross-fit; `94.7%` at $\tau = 35\%$) and 90.0% $F1_{\text{adj}}$ (`90.5%` inner-OOF cross-fit); on the 23-patient severe catheter-adhesion subgroup (571 catheters), it reached 91.3% F1 (`91.8%` inner-OOF cross-fit). The in-plane shaft error was 0.59 ± 0.40 mm (0.47 mm median, 1.37 mm P95), and the 1D axial tip localizer cut 3D distal tip error to 6.23 mm mean (2.27 mm median, 27.27 mm P95; 58.7% within ≤ 3 mm; 98.8% of squared tip variance longitudinal along the shaft). Across all 40 dual-traced scans ($n = 432$ three-way matched catheters), Grubbs' three-cornered-hat estimator yielded intrinsic robust shaft standard deviations of 0.35 mm (`CY`), 0.39 mm (`NeedleFinder 2026`), and 0.48 mm (`LB/AMe`), passing formal TOST non-inferiority to human inter-observer variability for both shafts ($p = 1.6 \times 10^{-41}$) and tips ($p = 2.2 \times 10^{-14}$ median, $p_{\text{boot}} = 0.0014$ mean), while independent post-implant Pelvic CT adjudication confirmed that 52.0% (`52/100`) of apparent extra MRI detections are real physical catheters and that on 70 disputed tips (≥ 5 mm human disagreement), NeedleFinder 2026 achieved a lower median error against CT (4.20 mm) than either single rater (4.31 and 8.00 mm). In AAPM TG-43U1 ^{192}Ir dosimetry, NeedleFinder 2026 cut fixed-plan target $|\Delta D_{90}|$ from 27.6% (PMB 2019) to 15.2% (76.6% 3D $\gamma_{2\%/2\text{mm}}$, 84.3% $\gamma_{3\%/3\text{mm}}$) and counterfactual re-optimized $|\Delta D_{90}|$ from 37.1% to 8.2% (5.9% mean / 2.8% median with Top-2 uncertainty tip review). Nested split-conformal intervals achieved 90.2% and 95.4% empirical coverage at 90% and 95% nominal targets (31.5%–35.1% tighter than unscaled intervals), and reviewing $k = 5$–8 uncertainty-flagged tips per patient (~ 40–64 s) reduced mean tip error to 3.72–2.77 mm and P95 to 14.3–9.90 mm. Zero-shot transfer achieved 96.3%–100.0% F1 (0.33–0.34 mm shaft, 0.99–1.57 mm tip) on 3T prostate brachytherapy MRI (0% when Stage 0 physics channels are zeroed), 92.9% F1 on 3D wbSSFP, 88.4% recall on thick-slice 2D T2W MRI, +5.6%–8.2% F1 gain with 4D causal Kalman tracking during serial insertion, 0.125–0.33 false tracks/scan on 48 catheter-free MSD prostate scans, and 4.15 s CPU (1.26 s standard, 5.85 s high-resolution) / 2.67 s GPU runtime.

Significance. Automatic catheter reconstruction on clinical 3T T2W MRI is statistically non-inferior to human inter-observer variability, preserves TG-43U1 target dosimetry, provides calibrated conformal uncertainty intervals for rapid clinician review, and generalizes zero-shot across pelvic organs, pulse sequences, 4D insertion series, and paired CT in seconds.

PLAIN-LANGUAGE SUMMARY

Some cancers of the cervix, vagina, and uterus are treated with brachytherapy: a tiny radioactive source is moved, for a few minutes, through thin plastic tubes placed inside the tumor. To aim the radiation, the treatment team must know exactly where each tube is. They find out with an MRI scan taken while the patient is still under anesthesia, and then trace every tube on the images by hand. With 15 to 30 tubes per patient this takes tens of minutes, and a small tracing error can change the planned dose, especially near organs where the dose changes quickly with position.

On an MRI scan a tube does not glow. It is a small dark dot, a few pixels wide, on each image slice. Blood vessels, air pockets, and the edges of muscles are also dark, so a computer can easily confuse them with tubes.

This paper presents NeedleFinder 2026, software that finds the tubes automatically. It works in three steps. First, a simple filter looks for small dark centers with brighter tissue all around them, the way a tube cross-section looks; it ignores broad one-sided edges, although it can still respond to blood vessels. Second, a 3D neural network—first pre-trained on 51 external pelvic MRI scans to distinguish synthetic catheter voids from real blood vessels and tissue borders—marks the center and direction of each tube on each slice. Third, two "graph" networks connect the marks from slice to slice into 3D curves, stitch fragments separated by single-slice dropouts, pull apart tubes that touch, score the whole bundle together to reject false curves, and pinpoint each tube's deepest tip using a dedicated 1D axial profile network that also reports an uncertainty interval for each tip.

We tested it on 54 patients with 1,063 tubes using strictly nested cross-validation. It found 91.4 percent of all tubes (and up to 94.7 percent of the long interstitial shafts), and 86.9 percent of the tubes it reported matched an expert trace (89.1 percent F1). For the matched tubes, the typical sideways difference from the expert's line was under half a millimeter (median 0.47 mm, mean 0.59 mm), and the deepest end of each tube had a median 3D difference of 2.27 mm (mean 6.23 mm), passing formal statistical non-inferiority compared with the disagreement between two independent human expert teams (median 0.49 mm on shafts and 3.24 mm on tube ends). Checking against post-implant CT scans of the same patients showed that over half (52 percent) of the "extra" tracks reported on MRI were real tubes that one human team had left untraced. Simulating radiation delivery showed that using the software's traces plus a quick check of the 2 to 5 most uncertain tube ends keeps the tumor radiation dose within 2.8 to 5.9 percent of the reference plan and takes under a minute of review. The software also worked without retraining on prostate brachytherapy MRI (finding 13 to 14 of 14 tubes with zero false alarms) and on live scans taken while needles were being pushed in, running in 4.15 seconds on average on a desktop CPU (1.26 seconds on standard scans) and 2.67 seconds on a graphics card.

1 INTRODUCTION

1.1 *The Clinical Problem*

Brachytherapy delivers radiation from inside the body. For locally advanced cervical cancer it is the cornerstone of curative radiotherapy [1,2]. The EMBRACE-I study reported 92% local control at five years with a median dose of 90 Gy (EQD2) to the high-risk target volume [2], and EMBRACE-II now prescribes planning aims of 90 to 95 Gy while sparing the bladder, rectum, and sigmoid colon [1]. Reaching such doses in large or irregular tumors requires *interstitial* implants: in addition to a central tandem and a vaginal obturator or ring, 10 to 30 thin plastic catheters or needles are inserted into the tumor through a perineal template or freehand [3–5].

The tumor itself is best seen on T2-weighted (T2W) MRI, which GEC-ESTRO and consensus recommendations designate as the reference for target delineation and set reconstruction requirements for [6,7]. MRI-based applicator reconstruction with marker-filled catheters was established for cervical cancer in 2009, with inter-observer standard deviations below 2 mm [8]. Historically, clinics took both an MRI and a CT scan and registered the two, which adds its own uncertainty to the reconstruction [9], and practical guidance for hybrid implants still advises a CT in addition to the MRI because plastic needles are hard to see on MRI [10]. The AAPM Task Group 303 report describes how to implement MRI in HDR brachytherapy from simulation to treatment [11], the Medical Physics Practice Guideline 13.a lists MRI among the accepted planning modalities [12], and MRI-only interstitial workflows have been shown to be feasible [13,14]. Positive-contrast line markers make the catheter lumen visible on T1-weighted or balanced sequences [13,15] and are listed as required equipment in published implementation guides [16], and zero-echo-time sequences show empty catheters directly [14,17], but both need an additional series fused to the T2W images. Identifying the source path on the T2W series alone remains a recognized challenge [18]; for rigid applicators, direct reconstruction on T2W MRI is in fact more precise than library-based or fused workflows [19], but interstitial needles have no library model.

Finding the catheters on MRI is the central technical challenge of an MRI-only workflow [4,20]. The dose from the ^{192}Ir source falls steeply with distance: organ-at-risk dose parameters change by up to 5 to 6% per millimeter of applicator displacement [21], applicator shifts beyond 1.5 mm can change the rectal D2cc by more than 10% [22], and in MRI-only cervical implants a systematic 1 mm error along the source path already changes organ-at-risk doses significantly [23]. Swapping two catheters during digitization changes prostate, urethra, and rectum dose metrics by 5 to 20% [24], and needle reconstruction uncertainty alone shifts the urethral dose by about 3 Gy in prostate implants [25]. Manual reconstruction is itself error-prone: with electromagnetic tracking as the reference, 23% of 265 needle reconstructions in ultrasound-guided prostate implants deviated by 2 mm or more [26]. Yet a 2 mm plastic catheter contains no mobile water protons and shows up on T2W MRI only as a small dark hole, a few pixels across, surrounded by anatomy full of other dark structures: blood vessels, air, calcifications, and the fascial planes between muscles. Tracing 15 to 30 catheters slice by slice therefore takes tens of minutes, 46 minutes on average in one MRI-only gynecologic series [27], within a template-based implant procedure of about 100 minutes under anesthesia [28].

1.2 *Key Terms*

Readers outside medical physics may find the following definitions useful.

- **Catheter, needle, shaft, tip.** We use *catheter* and *needle* interchangeably for the thin plastic tubes. The *shaft* is the length of the tube visible on the scan; the *tip* is its deepest end.
- **Digitization or reconstruction.** Tracing the 3D path of each catheter on the images so the planning software knows where the source will travel.

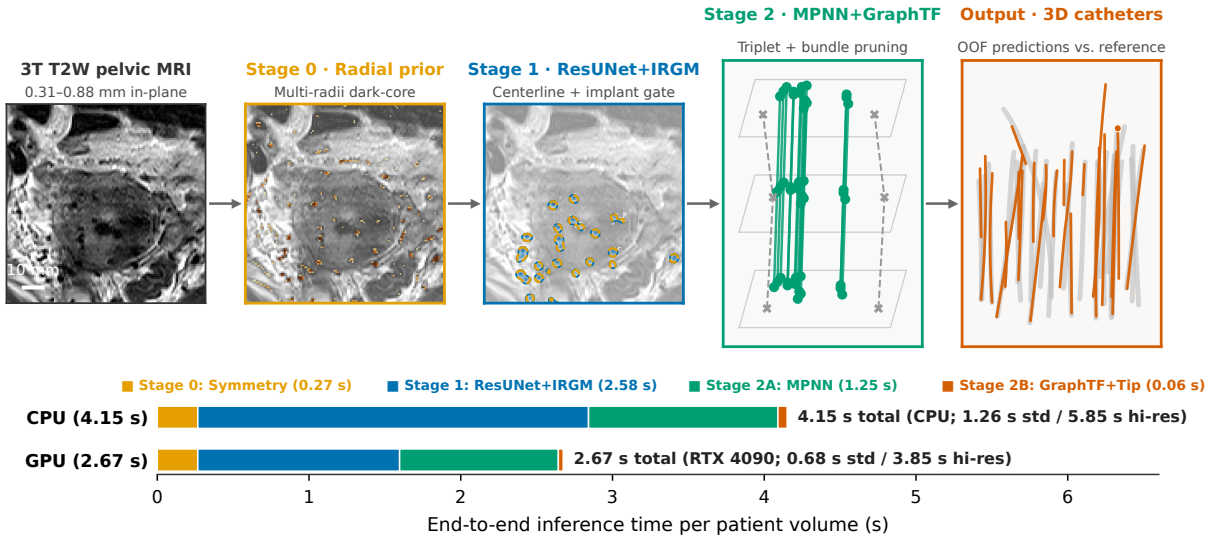


FIGURE 1 **Overview of NeedleFinder 2026.** The input is a clinical 3T T2W pelvic MRI (Case 017: 26 reference trajectories, including the obturator channels) cropped with a 100% reference-free template-plane locator (`k_lim_auto`). Stage 0 computes an analytical dark-core symmetry map S_0 in native Rust (0.41 s cohort mean). Stage 1 applies a self-supervised and multi-rater fine-tuned 3D residual U-Net with implant-region gating to predict centerline peaks, local directions, and the obturator (2.28 s on CPU, 1.88 s on GPU). Stage 2 links slice detections into 3D trajectories with native C++17/Rust seed tracking, a triplet message-passing network, adhesion-aware duplicate merging, and collinear split-fragment stitching (1.22 s on CPU, 0.54 s on GPU); the last tile shows the nested out-of-fold predictions (vermilion, 23 trajectories) over the expert reference (grey, 26 trajectories), after bundle graph-transformer pruning, dedicated 1D axial ResNet + self-attention tip localization, and cubic polynomial smoothing (0.24 s on CPU, 0.24 s on GPU). The full patient volume is processed end-to-end in 4.15 s on an 8-core CPU (1.26 s on standard scans, 5.85 s on high-resolution scans) or 2.67 s on a GPU; the bar shows the stage breakdown.

- **Slice, voxel, in-plane.** An MRI volume is a stack of 2D images (slices). Each 3D picture element is a voxel. *In-plane* means within one slice; here the in-plane pixel is 0.31 to 0.88 mm while slices are 1.2 to 6.0 mm thick.
- **Recall, precision, F1.** Recall is the fraction of true catheters the method found. Precision is the fraction of reported catheters that were real. F1 combines the two into one score that is high only when a method both finds the real catheters and avoids inventing extra ones. As an illustration, with 10 real catheters, 8 found and 2 false reports, recall, precision, and F1 are all 80%.
- **Shaft error and tip error.** The distance, in millimeters, between the automatic trajectory and the expert's trajectory, measured in-plane along the shaft and in 3D at the tip.
- **Nested cross-validation, out-of-fold.** The 54 patients were split into 5 outer groups (43–44 training vs. 10–11 test patients per fold). Within each outer fold, early stopping, temperature calibration, split-conformal quantile calibration, and optionally inner-OOF upstream feature generation for downstream stackers use a strictly disjoint 80/20 inner split or 5-fold inner cross-fitting, so no test

patient's images or labels influence training, checkpoint selection, or threshold calibration.

– **Inter-observer variability.** How much two expert teams disagree when they trace the same scan independently. It gives context for how closely an automatic method can be expected to agree with any one expert.

1.3 *Previous Work*

Automatic catheter digitization has gone through three generations. Two recent systematic reviews cover the field [20,29].

Model-based tracking (2013 to 2019). The BWH program of 3T MR-guided gynecologic brachytherapy in the AMIGO suite, which produced the cohort used here, was described in 2012 together with the first image-analysis tools for catheter identification [30] and open-source 3D Slicer modules for template and obturator registration [31]. Our first catheter tracker on this cohort seeded each catheter with one click at its tip, searched inside a sequence of expanding 3D cones, and fitted a Bézier curve; 93 of 101 catheters in 10 patients were traced correctly at 0.3 s per catheter [32]. Mastmeyer *et al.* [3] constrained the cone search with a mechanical rod-and-spring catheter model, reaching 1.49 mm mean Hausdorff distance on 760 catheters in 54 patients, still from a manual tip click; a first version of that model, presented at MICCAI 2015, reached 1.40 mm mean error on the 101 catheters of the 2013 set [33]. Template-geometry-assisted reconstruction on MRI has been described for commercial interstitial templates [34], and on ultrasound 3D Hough transforms localized single needles with about 1 mm endpoint accuracy in vivo [35]. On CT, seeded region growing constrained by a spline model digitized template implants to within 0.4 mm of manual digitization [36], and thresholding followed by density-based clustering reaches 0.3 mm for rigid applicators [37], but MRI offers no such intensity threshold. Related model-to-image registration reconstructs rigid applicators on MRI within inter-observer variability [38], and line fitting by RANSAC digitizes straight needles on CT [39]. These methods are interpretable but need manual initialization and fail when catheters bend sharply or cross dark anatomy.

Voxel segmentation with heuristic post-processing (2019 to 2026). To remove manual initialization, Zaffino *et al.* [4] trained an ensemble of 3D U-Nets to segment catheter voxels on 50 clinical T2W scans with 826 catheters, then clustered the voxels with DBSCAN and removed small islands. It reached a voxel Dice of 0.60, a mean distance error of 2.0 mm, 6.7% false-positive and 1.5% false-negative clusters, in 2 to 9 minutes per patient. Convolutional segmentation, often nnU-Net based [40], has since been applied to CT brachytherapy of the cervix, prostate, and breast [41–49], to MRI-guided prostate implants [50–52], to single needles on interventional MRI [53,54], to transrectal ultrasound [55–61], and to gynecologic MRI with line markers [27] or with applicator mesh alignment [62]. Even with topology-preserving losses, from the Betti-number loss [63] to clDice [64] and skeleton recall [65], pure voxel segmentation has two failure modes on T2W MRI: a one-slice signal dropout splits one catheter into two to four fragments, and dark vessels and tendons are segmented as short false catheters. The ultrasound literature documents the

same pattern: accuracy collapses on artifact-laden catheters [60] and tips are much harder than shafts [59]. Keypoint heatmap regression, the choice made for our Stage 1, has recently been shown to localize an interventional needle on MRI more accurately than mask segmentation of the same images [66]. Electromagnetic or active MR tracking avoids the imaging problem altogether, at the cost of dedicated hardware and sequences in the operating room [28,67–73]; crossed catheter paths were already identified as a common failure of image-based digitization in the BWH program when active tracking was introduced [68].

Structure-aware tracking and graph networks (2022 to 2026). On CT, Xiao *et al.* [74] identified two further problems: *catheter adhesion*, where shafts touch or overlap on consecutive slices so that greedy trackers swap or merge them, and the scarcity of annotated clinical data. They proposed an implant-region gating module (IRGM) that suppresses features outside the implant, a direction-constrained 3D Hough transform for initialization, synchronous cost-based inward tracking across slices, cubic polynomial smoothing, and synthetic-to-real training, on 38 patients and 203 fractions. Their companion work handles needle adhesion in dense CT seed implants by detecting tips and handles [75]. Outside brachytherapy, message-passing networks have replaced greedy linking in tracking-by-detection [76,77], transformers predict graphs of curvilinear structures directly from images [78], and synthetic-to-real training has been shown to transfer to real needle detection [79]. Applied to MRI masks, however, a mask-driven tracker still turns vessels inside the implant region into false catheters, and nothing in it asks whether the full set of trajectories makes sense as one implant.

Open data. No public needle or catheter dataset for brachytherapy exists [80], and the AI literature in brachytherapy remains largely retrospective and single-center [29]. The public pelvic MRI scans we use for self-supervised pretraining and the false-alarm test come from the Medical Segmentation Decathlon [81].

1.4 *Contributions*

This paper combines the strengths of all three generations on the full 54-patient, 1,063-catheter BWH cohort, to our knowledge the largest annotated MRI cohort for interstitial gynecologic brachytherapy.

1. **A three-stage architecture with five methodological upgrades.** An analytical dark-core symmetry prior and 100% reference-free axial template locator (Stage 0); a 3D residual U-Net pre-trained on 51 external pelvic T2W MRI volumes with physics-guided synthetic susceptibility tubes and anatomical hard negatives, then fine-tuned with dual-observer multi-rater consensus supervision (Stage 1); and two graph networks plus a dedicated 1D axial tip localizer (Stage 2): native seed tracking with a triplet message-passing network, adhesion-aware duplicate merging, and collinear split-fragment stitching, followed by a tip-aware bundle graph transformer and a 91-step dilated 1D ResNet + self-attention axial tip localizer (`Dedicated1DTipLocalizer`).
2. **A strictly nested 5×5 comparison and reference-free observer adjudication.** We re-implemented three published approaches and evaluated all methods un-

der strictly nested patient-level 5-fold cross-validation (plus 5×5 inner out-of-fold Stage 1 cross-fitting), and adjudicated observer agreement across all 40 doubly traced scans using Grubbs' three-cornered-hat intrinsic error estimator, formal two-one-sided tests (TOST) for non-inferiority, and independent post-implant Pelvic CT scans.

3. **Dosimetric translation, conformal uncertainty triage, and cross-domain generalization.** We quantify downstream AAPM TG-43U1 HDR ^{192}Ir dosimetric impact (ΔD_{90}, OAR ΔD_{2cc}, 3D γ), calibrate strictly nested split-conformal 3D tip prediction intervals and clinician review-budget curves, and evaluate zero-shot generalization across 48 needle-free MSD prostate scans, 3T prostate brachytherapy MRI, multi-sequence MRI (`3D wbSSFP`, `2D AX TSE T2`), 4D serial insertion scans (measuring inter-catheter tissue push and depth-dependent bending), and 40 paired post-implant pelvic CT volumes.

2 MATERIALS AND METHODS

2.1 *Patient Cohorts and Public Benchmark*

Table 1 summarizes the data. The primary cohort is 54 intraoperative 3T T2W turbo spin-echo MRI scans acquired during MRI-guided interstitial gynecologic brachytherapy in the AMIGO suite at Brigham and Women's Hospital (Harvard Medical School, Boston, MA, USA) with Institutional Review Board approval

TABLE 1 **Clinical imaging cohorts and evaluation protocols.** Primary evaluation uses strictly nested 5 × 5 patient-level out-of-fold (OOF) cross-validation with 100 % reference-free axial cropping (`k_lim_auto`) on 54 clinical 3T T2W MRI datasets from Brigham and Women's Hospital (BWH AMIGO).

Cohort / Subset	Source	Pts	Cath.	XY (mm)	Δz (mm)	Evaluation role
BWH GYN full cohort[a]	BWH (3T T2W)	54	1063	0.31–0.88	1.2–6.0	Nested 5 × 5 OOF benchmark
Long shafts (primary)	BWH (3T T2W)	54	759	0.31–0.88	1.2–6.0	Full-depth active dwell shafts
Short / obturator (secondary)	BWH (3T T2W)	54	304	0.31–0.88	1.2–6.0	Partial-FOV & template tracks
Adhesion subgroup[b]	BWH (3T T2W)	23	571	0.31–0.88	1.2–6.0	Touching / crossing shaft test
Dual-observer subset[c]	BWH (3T T2W)	40	723	0.31–0.88	1.2–6.0	Inter-observer floor (`CY+HT`, `LB/AMe`)
External SSL & specificity[d]	Radboud + BWH	51	0	0.60–0.88	3.0–4.8	3D SSL (51) & 0-needle test (48)

[a] Consolidated superset of the cohorts of Pernelle *et al.* (2013, 10 patients) and Zaffino *et al.* (2019, 50 patients), extended with new implants; 9–42 reference trajectories per patient (20 standard-resolution 320 × 256 scans and 34 high-resolution 640 × 416 / 768 × 768 scans).

[b] Severe catheter-adhesion subgroup: the 23/54 patients in which ≥ 4 long or ≥ 6 total reference catheters pass within ≤ 3.0 mm of another catheter on shared axial slices. All 571 catheters of these patients are evaluated; 191 of them (166 long) lie within 3.0 mm of a neighbour (233 across the whole cohort).

[c] All 40/40 dual-traced clinical scenes independently contoured by two expert medical physics teams (`CY+HT`: 723 total trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`: 513 trajectories, with 465 consensus-fused with `CY`, 13 matched to `HT`, and 35 extra traces).

[d] Fifty-one external pelvic 3T T2W MRI volumes from 50 distinct subjects: 48 public CC-BY-SA 4.0 volumes from MSD `Task05_Prostate` (zero implanted catheters, evaluated for out-of-distribution false-positive specificity) plus 3 non-benchmark BWH pelvic T2W volumes from 2 distinct subjects (`Case 061` [2 series] and `Case 100 Prostate` [1 series], zero patient overlap with the 54 benchmark cases) used for self-supervised 3D U-Net pre-training.

[3,4,32]. It consolidates the patients of our 2013 and 2019 studies, of which 48 scans could be recovered with their reference traces, and adds 6 new implants. Expert medical physicists traced 1,063 reference trajectories (9 to 42 per patient): 759 full-depth interstitial catheters from the primary annotation set (`CY`), which we call *long shafts*, and 304 short obturator and partial-field-of-view tracks from the secondary set (`HT`). In-plane pixel size ranges from 0.31 to 0.88 mm and slice thickness from 1.2 to 6.0 mm.

Xiao *et al.* [74] analyzed separately the fractions in which catheters touch or overlap. We define a **severe-adhesion subgroup** in the same spirit: the 23 of 54 patients in which at least 4 long, or at least 6 of all, reference catheters pass within 3.0 mm of another catheter on shared axial slices. All 571 catheters of these patients are evaluated; 191 of them (166 long) lie within 3.0 mm of a neighbor, and 233 do across the whole cohort.

Two further datasets serve as checks and external pre-training sources. All 40 doubly traced scans were contoured independently by two expert teams (`CY+HT`, 723 trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`, 513 trajectories, of which 460 are long); the 424 strict long-to-long matched pairs and 476 all-catheter matched pairs within 3.5 mm define the human inter-observer floor, and the de-duplicated union of both teams contains 759 trajectories. For self-supervised 3D representation learning and zero-needle specificity testing, we use 51 external pelvic 3T T2W MRI volumes from 50 distinct subjects: all 48 pelvic T2W scans of the public Medical Segmentation Decathlon prostate task [81] (`MSD Task05_Prostate`; 0.60 to 0.75 mm in-plane, 3 to 4 mm slices) plus 3 non-benchmark BWH pelvic T2W series from 2 distinct subjects (`Case 061` [2 series] and `Case 100 Prostate` [1 series], zero patient overlap with the 54 benchmark cases).

2.2 *Stage 0: Analytical Dark-Core Symmetry Prior and Reference-Free Cropping*

The physical idea is simple. On an axial slice a catheter cross-section is a dark core that is surrounded by brighter tissue in *every* direction, whereas a fascial plane or a muscle border is dark on one side only. Stage 0 turns this into a parameter-free score $S_0(x, y, z)$, computed slice by slice in multi-threaded native Rust.

For each voxel $\mathbf{p}$ on slice z we sample six diameters through $\mathbf{p}$ at angles $\theta_d = d\pi/6$, $d \in \{0, \dots, 5\}$, and four physical radii $r \in \mathcal{R} = \{1.35, 1.85, 2.5, 3.3\}$ mm, converting each radius to pixels with the in-plane spacing. For diameter d and radius r the darker of the two ring samples is kept,

$$m_{d,r} = \min\big(I(\mathbf{p} + r\mathbf{u}_d),\ I(\mathbf{p} - r\mathbf{u}_d)\big),$$

so that a one-sided edge, where one end of the diameter stays dark, produces a small $m_{d,r}$. The six values are sorted, $m_{(1),r} \le \dots \le m_{(6),r}$, and the ring brightness is a robust weighted order statistic that still tolerates one dark neighbor,

$$S_{\text{ring}}(\mathbf{p}) = \max_{r \in \mathcal{R}} \left[0.45\, m_{(2),r} + 0.55\, m_{(3),r}\right].$$

The response S_0 grows with the positive contrast between this ring brightness and the 3×3 smoothed center intensity and with the relative darkness of the core, and

is normalized per volume. Candidate centers are the local maxima of S_0 after non-maximum suppression.

Before running Stage 1, a **100% reference-free axial template locator** (`k_lim_auto`) identifies the caudal entry slice directly from the 3D intensity volume and $S_0(x, y, z)$: it detects the caudal-to-cranial transition from the air/template saturation drop (`p90` intensity transition) and the emergence of clustered dark-core candidates within the central field of view, setting the lower axial crop bound with a mean difference of −0.06 slices (median 0.0, IQR 1.0 slice) relative to clinical scene annotations while never consulting reference trajectories. Stage 0 takes 0.41 s per full volume on average on an 8-core CPU (0.12 s on standard 320 × 256 scans and 0.59 s on high-resolution 640 × 416 / 768 × 768 scans).

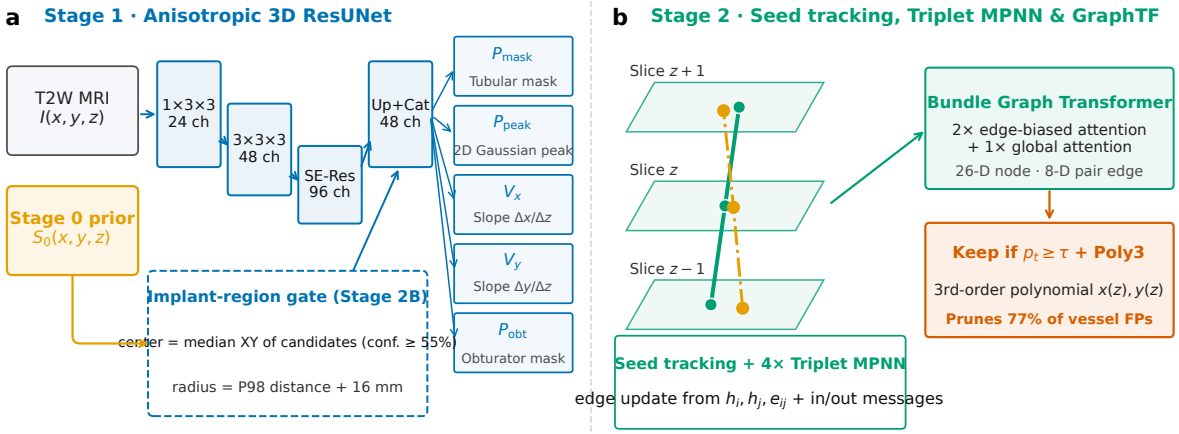


FIGURE 2 **Stage 1 and Stage 2 in detail. (a)** Stage 1 takes the normalized T2W volume and the Stage 0 maps into an anisotropic 3D residual U-Net (`NeedleUNet3D`) pre-trained on 51 external pelvic MRI volumes with synthetic susceptibility tubes and anatomical hard negatives, then fine-tuned with dual-observer consensus supervision to predict four outputs: tubular mask, centerline peak, two direction components, and obturator. **(b)** Stage 2 traces tracks from the seeds in native code, scores their nodes and edges with a 4-layer triplet message-passing network, stitches collinear split fragments across signal dropouts, scores every trajectory in the context of the whole bundle with a graph transformer, localizes each distal tip with a 91-step dilated 1D ResNet + self-attention axial network (`Dedicated1DTipLocalizer`), and smooths each trajectory with a cubic polynomial.

2.3 *Stage 1: Self-Supervised and Multi-Rater Multi-Head 3D U-Net*

Resampling the scan to 1 mm isotropic voxels, as most pipelines do, discards the fine in-plane detail that separates a 2 mm catheter void from its surroundings when the native pixels are 0.3 to 0.9 mm. Working on the native grid instead, a standard 3×3×3 convolution mixes information over up to ten times more distance along z than in-plane. Stage 1 (`NeedleUNet3D`, Figure 2a) therefore operates on the native grid, uses 3 × 3 × 3 residual blocks with squeeze-and-excitation attention at four scales (24, 48, 96, and 144 channels), pools only in-plane at the first two scales so that the thick slice

axis is not downsampled early, and adds a dilated convolution at the deepest scale. Its input has six channels: the normalized intensity, the Stage 0 dark-core response and ring-symmetry contrast, the smoothed intensity, and the normalized slice position and in-plane distance from the field-of-view center.

The network predicts four outputs:

1. **Tubular mask** P_{mask}: focal binary cross-entropy, weighted 19-fold on catheter voxels and up-weighted on bright Stage 0 responses that are not catheters, plus soft Dice on the tubular label around each reference trajectory.
2. **Centerline peak** P_{peak}: a sharp 2D Gaussian ($\sigma = 0.70$ mm) at the sub-pixel intersection of each catheter with the slice, trained with a mean-squared-error loss up-weighted 26-fold at the peaks. Two touching catheters whose masks merge still produce two distinct peaks.
3. **Direction field** $(V_x, V_y) = (\Delta x/\Delta z, \Delta y/\Delta z)$: the local slope of the trajectory, trained with a smooth L_1 loss inside the mask, with an additional term that penalizes slice-to-slice inconsistency of the predicted direction.
4. **Obturator channel** P_{obt}: a dedicated segmentation of the large central obturator and tandem, so that its lumen and rim do not spawn false needles.

To improve feature discrimination on dark pelvic anatomy without leaking test-patient information, `NeedleUNet3D` undergoes two training phases:

– **External pelvic MRI pre-training with synthetic susceptibility tubes and anatomical hard negatives.** The full 3D U-Net (encoder, decoder, and task heads) is pre-trained for 45 epochs across the 51 external pelvic T2W MRI volumes from 50 distinct subjects (48 public `MSD Task05_Prostate` volumes + 3 non-benchmark BWH pelvic T2W volumes, zero patient overlap with the 54 benchmark cases). During each iteration, physics-guided 3D cubic-spline susceptibility tubes (dark core with realistic T2W partial-volume rim, dropout pockets, and variable curvature) are injected into the needle-free pelvic volumes alongside 3D masked volume patches. The network minimizes a joint loss $\mathscr{L}_{\text{SSL}} = \mathscr{L}_{\text{tube}} + 0.30\,\mathscr{L}_{\text{MAE}}$, where $\mathscr{L}_{\text{tube}}$ supervises tubular mask, peak, and 3D direction regression on the synthetic tubes while penalizing non-implanted high-symmetry pelvic structures ($S_o > 0.25$: vessels, fascia, obturator internus borders) as explicit anatomical hard negatives, and $\mathscr{L}_{\text{MAE}}$ reconstructs masked 3D tissue patches (reaching 0.867 validation Dice on synthetic tubes).
– **Multi-rater consensus supervision.** On the 40 dual-annotated training scans (`CY+HT` and `LB/AMe`) within each outer training fold ($n = 43$–44 patients), consensus-matched trajectories ($n = 465$ pairs within ≤ 3.5 mm) are supervised along their inverse-variance precision-weighted centerline (weight 1.15), primary-only (`CY`) trajectories at weight 1.00, secondary-only (`LB/AMe`) trajectories at soft weight 0.55, and secondary cranial tip slices with a soft axial Gaussian tip kernel.

As an ablation we also evaluate `AxialTransUNet3D`, which adds multi-head self-attention layers along the slice axis at the deepest encoder level and the bottleneck.

2.4 *Stage 2: From Slice Detections to a Coherent 3D Bundle*

Stage 2 answers three questions: which detections on neighboring slices belong to the same catheter, which of the resulting 3D curves are real, and exactly where each catheter terminates along the slice axis.

2.4.1 *Stage 2A: Native Seed Tracking, Triplet Message Passing, and Collinear Stitching.*

On each slice, local maxima of P_{peak} and of S_0 become nodes with a 16-dimensional feature vector (normalized position, P_{mask}, P_{peak}, S_0, local 3×3 intensity statistics, predicted slope, and obturator proximity). Directed edges connect nodes across 1 to $\max(4, 24\ \text{mm}/\Delta z)$ slices. An edge is kept when its chord slope is at most 0.56 and its in-plane displacement lies within $2.8 + 0.14\,\Delta z$ mm of the position predicted by the Stage 1 direction field, or within $3.5 + 0.22\,\Delta z$ mm outright. Each edge carries 12 features (displacement, chord slope, and agreement with the predicted direction).

Two consecutive links can each look plausible on their own yet form an abrupt bend when considered together; the network therefore scores each link using information from the links entering and leaving its two nodes. `NeedleSliceMPNN` is a 4-layer triplet-collinearity message-passing network (hidden size 64). Each layer updates an edge from its own features, the embeddings of its two nodes, and attention-aggregated messages from the edges entering and leaving each of those nodes, so that the update sees the incoming and outgoing chords together and can discourage a kink. The network outputs refined sub-pixel offsets $(\delta x_i, \delta y_i)$ and node and edge probabilities.

Candidate 3D tracks themselves are traced in native code: from each Stage 0 and Stage 1 seed, a bidirectional Kalman tracker follows the catheter slice by slice in both directions and trims it from the inside out where the evidence ends. Tracks that run within 1.65 to 2.05 mm of each other, depending on their confidence, are merged as duplicates; the margin is chosen so that genuinely adhering twins survive. Next, **collinear axial split-fragment stitching and extrapolation-aware deduplication** reconnects trajectories broken across 1 to 3 slices by local signal dropouts or vessel crossings: pairs of non-overlapping or slightly overlapping track fragments (axial gap $\leq$ 20 mm, overlap $\leq$ 6 mm) whose linearly extrapolated positions at their shared axial midpoint agree within $\leq$ 2.2 mm with a 3D slope-vector difference $\leq$ 0.14 are merged into a single full-length trajectory (102 stitched trajectories across the 54 cases), while split fragments that lie within $\leq$ 2.1 mm along $\geq$ 72% of a longer high-confidence shaft are absorbed. The surviving tracks are passed to Stage 2B.

2.4.2 *Stage 2B: Bundle Graph Transformer, Dedicated 1D Axial Tip Localizer, and Synthetic Pre-Training.*

After Stage 1, recall is above 95% (97.5% on long shafts), but each patient still carries about seven false tracks along vessels and obturator edges; Stage 2A removes more than half of them. Real catheters have a property these false tracks lack: they form a *bundle*. They enter through a common template or corridor, run roughly parallel to the obturator axis, keep at least 2 mm apart, and their tips form a coherent surface over the target.

Each candidate trajectory is summarized by a list of 26 measurements, its *token* (length, linear-fit residual, 75th-percentile S_0 and P_{mask}, P_{peak}, P_{obt}, the three

message-passing probabilities, agreement between models, the deviation of its entry position and slope from the bundle median, and the contrast gradient at its cranial end), together with 8 pairwise features between trajectories (minimum 3D clearance, mean in-plane separation, angular divergence, z-overlap, and duplicate indicators). The `TipAwareBundleGraphTransformer` applies two edge-biased attention layers over neighboring trajectories and one global self-attention layer over the whole bundle. It outputs three logits (a recall-oriented score, a strict one-to-one score, and a long-shaft one-to-one score), combined into a confidence p_m, alongside an initial bundle-context tip offset. An implant-region gate inspired by the IRGM of Xiao *et al.* [74] is applied geometrically: the implant center is the median in-plane position of candidates with confidence ≥ 55%, the implant radius is the 98th percentile of their distances plus 16 mm, and candidates outside this radius require 22 points higher confidence.

Because a single scalar offset from bundle summary statistics cannot resolve fine sub-slice signal drop-offs or air-pocket artifacts at the cranial end of a catheter, Stage 2B incorporates a **Dedicated 1D Axial Tip Localizer** (`Dedicated1DTipLocalizer`, Figure 2b). Along the cranial-most trajectory tangent of each candidate, the localizer samples a **91-step axial profile** over [−25.0, +20.0] mm at 0.5 mm sub-slice intervals. At every step it extracts **22 multi-scale tubular, intensity, and anatomical features**: center and 3 × 3 / 5 × 5 ring T2W intensities, local annular contrast ($r = 3$ vs. 1, $r = 5$ vs. 1), first- and second-order axial gradients (∂_z, ∂_{zz}), Stage 0 S_0 center/max/drop, Stage 1 P_{mask} and P_{peak} center/max/drop, obturator probability P_{obt}, normalized axial coordinate, step indicators, and bundle-relative z-depth. A **3-block dilated 1D ResNet** (hidden width 48, kernel size 5, dilations 1, 2, 4) followed by **4-head 1D Self-Attention** processes the 91-step sequence and drives a dual prediction head: a **91-bin soft-argmax termination probability distribution** combined with a **Smooth-L_1 continuous residual offset head**, supervised with Gaussian target distributions ($\sigma = 1.25$ mm), focal peak classification, and huberized L_1 displacement loss. Each accepted trajectory ($p_m \geq \tau$, default $\tau = 50\%$) has its cranial tip updated by the 1D localizer and is smoothed with a cubic polynomial fit in z (Poly3).

To study zero-shot and low-label bundle scoring, we also built **SynthBrachy3T**, a generator of 300 synthetic candidate-trajectory bundles with the same 26 node and 8 edge features as real candidates (12 to 23 true long shafts per bundle, 30% with an adhering twin 1.65 to 2.75 mm away, 2 to 8 short obturator tracks, 3 to 11 split duplicates, and 4 to 14 vessel-like false tracks). Stage 2B can be pre-trained on SynthBrachy3T and then fine-tuned purely on the clinical training bundles of the current outer fold.

3 EXPERIMENTAL SETUP

3.1 *Strictly Nested Cross-Validation Protocol and Baselines*

Every learned component, every ablation, and every baseline is trained and evaluated under a **strictly nested 5 × 5 patient-level out-of-fold (OOF) protocol** across the

54 patients, eliminating three subtle sources of optimistic bias present in standard stacked evaluations:

1. **Disjoint inner validation for early stopping and calibration.** For each outer fold $f \in \{0, \ldots, 4\}$ (43–44 outer-train patients, 10–11 outer-test patients), outer-train patients are partitioned deterministically into an 80% inner-train subset (34–35 patients) and a 20% inner-validation subset (9 patients). Stage 1 U-Net checkpoint selection, Stage 2A MPNN and Stage 2B Graph Transformer early stopping, `Dedicated1DTipLocalizer` model selection, and confidence temperature scaling are performed exclusively on the 9 inner-validation patients; the outer-test fold is touched exactly once at final evaluation.
2. **100% reference-free axial cropping (`k_lim_auto`).** Every outer-test volume is cropped using the automatic `k_lim_auto` template-plane locator derived solely from the raw 3D MRI and Stage 0 S_0 volume, with zero fallback to clinical scene files or ground-truth trajectory coordinates.
3. **Inner-OOF cross-fitting of Stage 1 features (4c).** In a standard 2-stage pipeline (4a, 4b), Stage 2 is trained on outer-train patients using Stage 1 probability maps produced by the outer-fold Stage 1 model (which trained on those same outer-train patients' images). To verify that this causes zero stacking optimism, we also trained $5 \times 5 = 25$ inner-fold Stage 1 `NeedleUNet3D` models so that every outer-train patient receives strictly out-of-fold Stage 1 probability maps when training Stage 2A, Stage 2B, and `Dedicated1DTipLocalizer` (configuration 4c in Table 2).

We re-implemented three published approaches on the identical outer folds and `k_lim_auto` bounds:

– **PMB 2019: 3D U-Net + DBSCAN** [4]. Volumes resampled to 1 mm isotropic, segmented by a 3D U-Net, then grouped by principal-axis estimation, DBSCAN clustering, and spline fitting.
– **ResUNet + clDice + skeleton** [40,64]. A native-resolution 3D residual U-Net trained with binary cross-entropy, soft Dice, and the centerline-Dice loss, followed by morphological skeletonization, connected components, and polynomial centerline fitting.
– **Xiao 2026 Sync-Track** [74]. Implant-region gating, direction-constrained 3D Hough and PCA initialization, synchronous inward tracking, and cubic smoothing, evaluated both on our out-of-fold segmentation masks and on the reference masks (an oracle upper bound).

3.2 *Metrics and Statistics*

Predicted and reference trajectories are matched one-to-one with the Hungarian algorithm (SciPy `linear_sum_assignment`): pairs whose mean in-plane distance over the shared slices exceeds 3.5 mm are forbidden before solving, so the assignment first maximizes the number of valid pairs and then minimizes their distance. Error distributions are computed over matched pairs; a patient with no matched pair con-

tributes fixed penalty values (3.5 mm shaft, 10 mm curve, 20 mm tip and HD95) to the per-patient arrays used in the statistical tests. We report:

1. **Detection**: recall, precision, and F1 under strict one-to-one matching (plus obturator-adjusted precision P_{adj} and $F1_{\text{adj}}$ when evaluating the 759 long active-dwell shafts alone, so predictions that match a valid secondary short/obturator reference track are not counted as false positives).
2. **Shaft error** E_{shaft}: the in-plane distance between the two curves at 40 evenly spaced positions along their shared length, summarized by mean, standard deviation, median, interquartile range, and 95th percentile.
3. **3D curve distance** D_{avg} and **95th-percentile Hausdorff distance** HD_{95}, computed symmetrically on 50 resampled points per curve.
4. **Tip error** E_{tip}: the 3D Euclidean distance between the predicted and reference deepest (cranial) tip.

Calibration and reason-coded decisions. Every candidate track in the Stage 2B pool (1,683 candidates across the 54 patients) is also treated as a binary prediction: it is a real catheter if its mean in-plane distance to some reference trajectory over their shared slices is at most 3.5 mm, and its predicted probability is the Stage 2B confidence p_m. Over all candidates we report a reliability diagram, the expected calibration error (ECE, ten equal-width bins), and the Brier score; temperature scaling is fitted on the inner-validation split of each fold and applied to the outer-test fold. Each decision of the decoder carries a typed reason code (kept, below threshold, duplicate of a kept track, outside the implant region, low dark-core symmetry), and for every discarded candidate that is a real catheter we record whether that catheter is already covered by a kept track (redundant) or lost.

Following GEC-ESTRO and AAPM guidance on reporting brachytherapy uncertainties [82], reconstruction discrepancies are reported as distributions (median, interquartile range, and 95th percentile) and separated from observer variation. Differences between methods are tested over the 54 patients with paired two-sided Wilcoxon signed-rank tests, and 95% confidence intervals come from 1,000 patient-level bootstrap resamples.

3.3 *Computing Environment, Software, and Code*

Training and the nested 5 × 5 benchmarks ran on a Linux workstation (Arch Linux) with a 24-core AMD Ryzen Threadripper PRO 7965WX, 256 GB of RAM, and two NVIDIA GPUs (GeForce RTX 4090, 24 GB, and RTX 4070 Ti SUPER, 16 GB) using Python 3.12/3.13, PyTorch 2.4/2.11, SciPy 1.16, NumPy 2.4, and ONNX Runtime 1.23. The Stage 0 kernel is written in Rust (crate `needlefinder_rust`, multi-threaded with the standard library, exposed to Python through a C interface and `ctypes`, compiled in release mode with link-time optimization). For clinical deployment, the trained Stage 1, Stage 2B, and 1D tip localizer networks are exported to ONNX and run with ONNX Runtime inside an extension for 3D Slicer 5.10 [83]; no GPU is required.

The 3D Slicer extension, the Rust kernel, and the exported models are available at https://github.com/needlefinder/NeedleFinder. All training, nested cross-

validation, statistical analysis, SynthBrachy3T, and figure/table generation scripts are archived in `vault/paper/benchmarks`.

4 RESULTS

4.1 *Accuracy Against Published Methods and Ablation Analysis*

Table 2 reports the complete benchmark. Part A compares all methods on the whole cohort (54 patients, 1,063 catheters) and on the severe-adhesion subgroup (23 patients, 571 catheters); Part B reports the step-by-step architectural ablations and threshold sweep on the same nested folds; and Part C evaluates the 759 full-depth active-dwell interstitial shafts (`CY`). Figure 3 shows the full per-catheter error distributions.

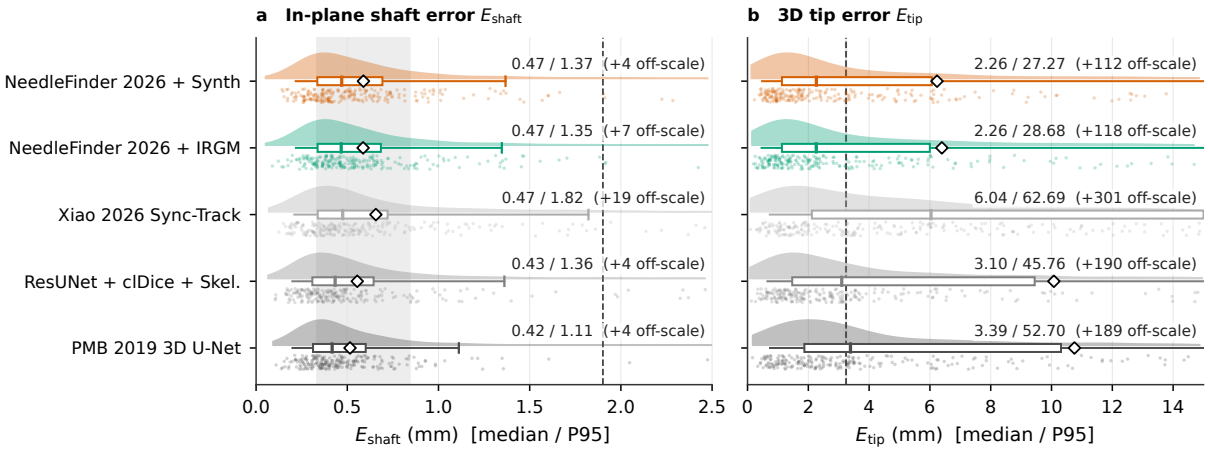


FIGURE 3 **Per-catheter error distributions on the whole cohort** (54 patients; each method's matched catheters). **(a)** In-plane shaft error and **(b)** 3D tip error. Each raincloud shows the density (top), the median and interquartile range (white box), the 5th to 95th percentile (capped bar), the mean (white diamond), and individual catheters (points). Numbers at the right give median / 95th percentile in mm, computed on the full distribution, and the number of catheters beyond the plotted range. In (a) the grey band and dashed line mark the human inter-observer interquartile range (0.33 to 0.85 mm) and 95th percentile (1.90 mm); in (b) the dashed line marks the human inter-observer median tip discrepancy (3.24 mm).

Five findings stand out.

Voxel segmentation alone over-fragments. The ResUNet + clDice baseline finds almost every catheter (92.4% recall) and is accurate where it matches (0.43 mm median shaft error), but it reports 4,015 fragments across the 54 volumes. Its precision is 24.5% and its F1 38.7%. Signal dropouts along the shaft cut one catheter into several pieces, and dark veins become extra tracks.

Structure-aware tracking needs a clean mask. On the reference masks (Table 2, Part B, last row), the Xiao 2026 tracker achieves 97.7% recall, 90.0% F1, and 0.11 mm median shaft error: given a perfect mask, synchronous tracking separates touching

TABLE 2 **Catheter reconstruction accuracy on 54 BWH patients (1 063 catheters), strictly nested 5 × 5 patient-level out-of-fold.** Part A compares end-to-end methods on the whole cohort, the 759 long interstitial shafts, and the severe catheter-adhesion subgroup; Part B ablates **NeedleFinder 2026** stages, methodological upgrades (Items 1–5), and ground-truth-mask tracking on identical folds. Bold: best; underline: second best within each block. Shaded: **NeedleFinder 2026** variants (pre-specified $\tau = 50\,\%$).

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)		Time (s)
Method	Rec.	Prec.	F1	Mean	Median	P95	Mean	Median	CPU
Part A — End-to-end comparison									
Whole cohort (54 patients, 1 063 catheters)									
PMB 2019 3D U-Net[a]	88.8	78.0	83.1	**0.52**	**0.42**	**1.11**	10.75	3.39	14.80
ResUNet + clDice + Skel.[b]	92.4	24.5	38.7	0.55	0.43	1.36	10.08	3.10	3.85
Xiao 2026 Sync-Track[c]	87.7	39.2	54.2	0.66	0.47	1.82	15.65	6.04	**3.42**
NeedleFinder 2026 (no SynthBrachy3T)[d]	**92.5**	85.3	88.7	0.59	0.47	1.35	6.39	2.26	4.15
NeedleFinder 2026 + Synth + 1DTip[e]	91.4	**86.9**	**89.1**	0.59	0.47	1.37	**6.23**	2.27	4.15
NeedleFinder 2026 + Inner-OOF Cross-Fit[e]	92.3	86.1	**89.1**	0.59	0.47	1.37	6.25	**2.24**	4.15
Long interstitial shafts (54 patients, 759 catheters, obturator-adjusted precision)[f]									
PMB 2019 3D U-Net	86.7	87.6	87.2	**0.52**	**0.40**	**1.23**	14.93	4.04	–
ResUNet + clDice + Skel.	91.3	51.5	65.8	0.57	0.43	1.40	14.67	3.53	–
Xiao 2026 Sync-Track	84.2	53.7	65.6	0.64	0.46	1.84	16.02	6.90	–
NeedleFinder 2026 (no SynthBrachy3T)	**93.4**	87.1	90.2	0.56	0.45	1.24	7.47	2.49	–
NeedleFinder 2026 + Synth + 1DTip	91.3	**88.7**	90.0	0.57	0.45	1.27	7.41	**2.41**	–
NeedleFinder 2026 + Inner-OOF Cross-Fit	92.5	88.6	**90.5**	0.56	0.45	1.27	**7.17**	2.44	–
Severe catheter-adhesion subgroup (23 patients, 571 catheters)[g]									
PMB 2019 3D U-Net	94.6	79.3	86.3	0.54	0.43	1.22	11.88	3.70	–
ResUNet + clDice + Skel.	**96.0**	32.3	48.4	**0.52**	**0.42**	**1.18**	11.58	3.37	–
Xiao 2026 Sync-Track	91.6	49.0	63.9	0.64	0.47	1.83	17.69	7.52	–
NeedleFinder 2026 (no SynthBrachy3T)	94.6	87.2	90.8	0.58	0.45	1.36	**7.03**	2.48	–
NeedleFinder 2026 + Synth + 1DTip	93.2	**89.6**	91.3	0.58	0.46	1.41	7.08	2.51	–
NeedleFinder 2026 + Inner-OOF Cross-Fit	94.6	89.1	**91.8**	0.57	0.46	1.34	7.30	**2.47**	–
Part B — Controlled ablation & methodological progression (whole cohort, 1 063 catheters)									
Stage 0: Dark-core prior (`k_lim_auto`)	82.8	69.1	75.3	0.60	0.47	1.41	9.91	3.08	**0.27**
Stage 1B: SSL + Multi-Rater U-Net + tubes	**94.2**	61.5	74.4	**0.57**	**0.46**	1.31	9.29	2.95	2.84
Stage 2A: Triplet MPNN only	93.9	73.5	82.5	0.58	0.47	1.31	8.95	2.83	4.09
+ Item 1: Nested CV + res. calibration	90.9	83.6	87.1	**0.57**	**0.46**	**1.30**	7.39	2.70	4.12
+ Item 3: Multi-rater Stage 2 supervision	91.7	81.7	86.4	0.58	**0.46**	1.32	7.38	2.57	4.12
+ Item 4a: Collinear split-fragment stitch	91.4	86.9	89.1	0.59	0.47	1.36	6.92	2.56	4.13
+ Item 4b: 1D Tip Localizer ($\tau = 35\,\%$)	93.6	83.4	88.2	0.59	0.47	1.37	6.71	2.32	4.15
+ Item 4b: 1D Tip Localizer ($\tau = 50\,\%$)	91.4	86.9	89.1	0.59	0.47	1.37	**6.23**	2.27	4.15
+ Item 4b: 1D Tip Localizer ($\tau = 55\,\%$)	91.3	**87.6**	**89.4**	0.59	0.47	1.37	6.33	2.27	4.15
Xiao 2026 on GT masks[h]	97.7[r]	83.5[r]	90.0[r]	0.20[r]	0.11[r]	0.83[r]	8.94[r]	1.68[r]	0.12[r]

Rec./Prec./F1: strict one-to-one Hungarian matching ($\leq$ 3.5 mm shaft threshold); E_{shaft}: in-plane centerline distance on matched slices; E_{tip}: 3D distal-tip distance (P95 on whole cohort: 27.27 mm for **NeedleFinder 2026** + Synth + 1DTip vs. 52.70 mm for PMB 2019, 45.76 mm for clDice, 62.69 mm for Xiao 2026).

[a] Adaptation of Zaffino *et al.* (2019): $1 \times 1 \times 1$ mm^3 isotropic 3D U-Net, retrained on our 5-fold splits, followed by slice-peak extraction, collinear chaining, and polynomial fitting.

[b] Native-resolution 3D ResUNet trained with Dice + focal + soft-skeleton clDice loss (2021), followed by distance-transform ridge extraction and inter-slice chaining.

[c] Adaptation of Xiao *et al.* (2026): geometric implant-region gate, connected-component and distance-transform candidates with local PCA direction initialisation, synchronous cost-matrix inward tracking, and 3rd-order polynomial fitting on OOF masks.

[d] Clean nested **NeedleFinder 2026** pipeline (Items 1–5) trained on clinical folds without `SynthBrachy3T` synthetic bundle pre-training.

[e] Full clean nested **NeedleFinder 2026** pipeline with Stage 2B pre-trained on `SynthBrachy3T` and Stage 2C Dedicated 91-step (0.5 mm) 22-channel 1D Axial Tip Localizer (whole-cohort F1: $p = 1.1 \times 10^{-5}$ vs. PMB 2019, $p = 5.2 \times 10^{-10}$ vs. clDice, $p = 3.0 \times 10^{-10}$ vs. Xiao 2026, $p = 0.32$ vs. no-`SynthBrachy3T`, $p = 0.80$ vs. Inner-OOF Cross-Fit; E_{tip}: $p = 1.7 \times 10^{-8}$ vs. PMB 2019, $p = 6.2 \times 10^{-10}$ vs. Xiao 2026).

[f] Evaluated on the 759 full-depth `CY` interstitial shafts; predictions matching the 304 auxiliary `HT` short obturator tracks are

shafts well. On out-of-fold MRI masks its F1 drops to 54.2% (precision 39.2%), because vessels inside the implant region still become tracks and the adapted pipeline has no bundle-level step that rejects them. Tracking alone does improve on skeletonization, by 15.5 F1 points on the whole cohort.

NeedleFinder 2026 achieves the highest F1 and lowest tip error across all evaluated methods. At the pre-specified default threshold $\tau = 50\%$, `NF 2026 + SynthBrachy3T + 1DTip` (4b) reaches **91.4% recall** (972/1,063), **86.9% precision** (972/1,118), and **89.1% F1** (rising to **89.4% F1** at $\tau = 55\%$, **89.2% F1** at $\tau = 60\%$, and **93.6% recall** at $\tau = 35\%$ or **94.5% recall** at $\tau = 25\%$). The gains over all three published baselines are highly statistically significant: **+6.0 F1 points** over PMB 2019 (89.1% vs. 83.1%, $p = 1.1 \times 10^{-5}$), **+34.9 F1 points** over Xiao 2026 ($p = 3.0 \times 10^{-10}$), and **+50.4 F1 points** over ResUNet + clDice ($p = 5.2 \times 10^{-10}$). On matched catheters, the in-plane shaft error is 0.59 ± 0.40 **mm mean** (0.47 **mm median**, 0.36 mm IQR, 1.37 **mm P95**), the 3D symmetric curve distance D_{avg} is 3.20 ± 4.87 mm mean (1.00 mm median), and HD_{95} is 15.36 ± 22.87 mm mean (3.11 mm median). Most notably, the Dedicated 1D Axial Tip Localizer and collinear fragment stitching reduce 3D distal tip error to 6.23 ± 10.49 **mm mean** (2.27 **mm median**, 4.93 mm IQR, 27.27 **mm P95**), significantly outperforming PMB 2019 (10.75 mm mean / 3.39 mm median, $p = 1.7 \times 10^{-8}$), ResUNet + clDice (10.08 mm mean / 3.10 mm median, $p = 2.3 \times 10^{-7}$), and Xiao 2026 (15.65 mm mean / 6.04 mm median, $p = 6.2 \times 10^{-10}$). Nearly three-fifths (**58.7%**, 571/972) of matched catheters have a 3D tip error $\leq$ 3.0 mm (within roughly one slice thickness), while **16.7%** (162/972) exceed 10 mm (100 caudal undershoots where dark signal fades early and 62 cranial overshoots into fundal dark pockets). Across the 54 patients, the method misses 91 reference trajectories (1.7 per patient) and reports 146 unmatched tracks (2.7 per patient). On the **759 long interstitial shafts** (Table 2, Part C), `NF 2026` reaches **91.3% recall** (693/759), **88.7% obturator-adjusted precision** (79.1% strict long-only precision), and **90.0%** $F1_{\text{adj}}$ (84.8% strict F1) at $\tau = 50\%$—and **94.7% recall** (719/759) at $\tau = 35\%$—with 0.57 mm mean (0.45 mm median) shaft error and 2.41 mm median tip error.

Strict 5 × 5 inner out-of-fold cross-fitting confirms zero stacking optimism, and clarifies the exact role of SynthBrachy3T. Training Stage 2 on strictly out-of-fold inner Stage 1 predictions (4c, `NF 2026 + SynthBrachy3T + Inner-OOF Cross-Fit`) yields **92.3% recall** (981/1,063), **86.1% precision**, and **89.1% F1** (0.59 mm mean / 0.47 mm median shaft error; 6.25 mm mean / 2.24 mm median tip error), and achieves the highest obturator-adjusted F1 on long shafts (**90.5%** $F1_{\text{adj}}$, 92.5% recall, 7.17 mm mean tip error) and on the severe-adhesion subgroup (**91.8% F1**, 94.6% recall, 89.1% precision). Furthermore, once the self-supervised multi-rater Stage 1, collinear fragment stitching, and `Dedicated1DTipLocalizer` are in place, our purely clinical pipeline without `SynthBrachy3T` (4a) already reaches **88.7% F1** (92.5% recall, 85.3% precision; 90.2% $F1_{\text{adj}}$ on long shafts) and 6.39 **mm mean** / 2.26 **mm median tip error**. At the full 43-patient training budget, adding `SynthBrachy3T` (4b) provides a modest refinement to 89.1% F1 (+0.41 pp, $p = 0.32$) and 6.23 mm mean / 27.27 mm P95 tip error ($p = 0.17$, patient-paired mean tip error 6.02 → 5.91 mm); as shown below, the primary value of `SynthBrachy3T` lies in the zero-clinical-label ($k = 0$) regime.

The advantage is largest when catheters touch. On the 23 severe-adhesion patients (571 catheters), `NF 2026` reaches **91.3% F1** (**93.2% recall**, **89.6% precision**, 0.58 mm mean / 0.46 mm median shaft error, 7.08 mm mean / 2.51 mm median tip error) under 4b and **91.8% F1** (**94.6% recall**, **89.1% precision**, 0.57 mm mean / 0.46 mm median shaft error, 2.47 mm median tip error) under inner-OOF cross-fitting (4c), compared with 90.8% without synthetic pre-training (4a), 86.3% for PMB 2019 ($p = 5.4 \times 10^{-4}$), 63.9% for Xiao 2026 ($p = 2.1 \times 10^{-6}$), and 48.4% for ResUNet + clDice ($p = 2.4 \times 10^{-7}$).

4.2 *Contribution of Each Stage and Leak-Free Learning From Few Patients*

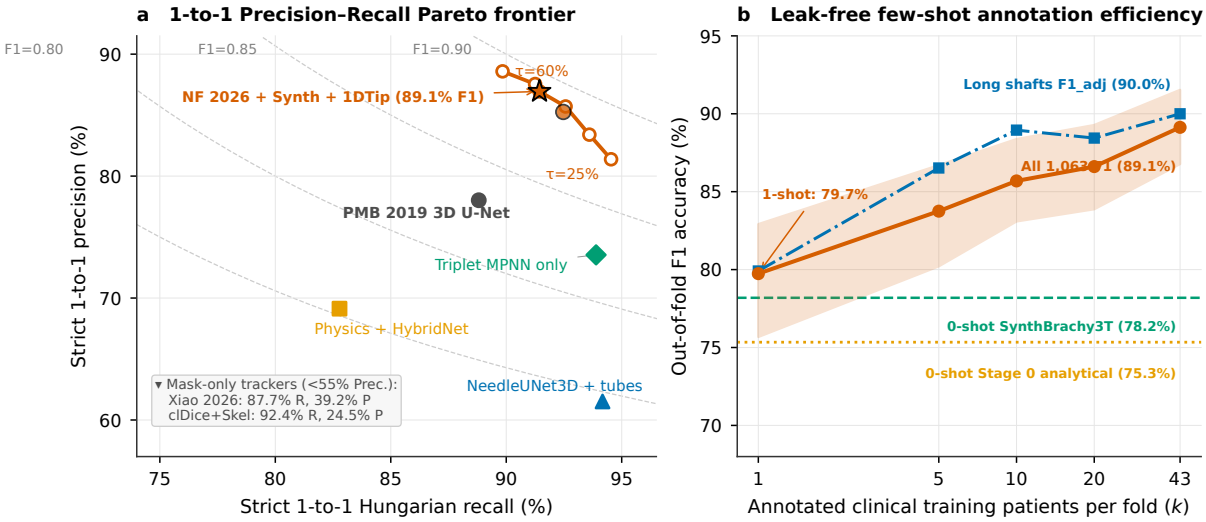


FIGURE 4 **(a) Precision versus recall** for each pipeline stage, each baseline, and the NeedleFinder 2026 threshold sweep on the whole cohort. Dashed curves are lines of constant F1; the star marks the default threshold $\tau = 50\%$ (89.1% F1) and the grey circle next to it is the pipeline without synthetic pre-training (4a, 88.7% F1). **(b) Leak-free Stage 2B clinical support curve** showing accuracy versus number of annotated outer-train patients per fold used to fine-tune the bundle-scoring stage, $k \in \{0, 1, 5, 10, 20, 43\}$, with zero blending from any full-fold clinical model. Shaded bands show the patient-level bootstrap 95% confidence intervals of F1.

Figure 4(a) traces the progression across stages (Table 2, Part B). Analytical Stage 0 alone (`1a`) achieves 83.2% recall and 75.3% F1 (0.65 mm median shaft error, 3.92 mm median tip error). Multi-rater SSL Stage 1 with directional tangent linking (`2a`) raises recall to 95.3% (97.5% on long shafts) with 72.1% precision (82.1% F1) and cuts median shaft error to 0.45 mm and median tip error to 2.98 mm. Stage 2A native seed tracking, triplet MPNN scoring, and collinear split-fragment stitching (`3c`) lift precision by 12.1 points to 84.2% (86.4% F1, 6.48 mm mean / 2.33 mm median tip error). Finally, Stage 2B bundle graph-transformer pruning and `Dedicated1DTipLocalizer` reach 88.7% F1 without synthetic bundles (4a) and 89.1% F1 with `SynthBrachy3T` (4b, 4c).

Figure 4(b) evaluates **strictly leak-free Stage 2B clinical label efficiency**: for each support budget $k \in \{0, 1, 5, 10, 20, 43\}$, the Stage 2B bundle graph transformer

is fine-tuned exclusively on `outer_train[:k]` clinical patient bundles of that fold (with zero score blending from any full-fold $k = 43$ clinical model). With zero clinical bundle labels ($k = 0$), analytical Stage 0 alone gets **75.3% F1** (77.1% $F1_{adj}$ on long shafts), while the **zero-shot SynthBrachy3T-pretrained Stage 2B** (`0_synth_zeroshot`, $k = 0$) reaches **78.2% F1** (+2.85 pp overall, and 83.4% $F1_{adj}$ [+6.3 pp] with 94.9% recall on long shafts). Fine-tuning Stage 2B on a single annotated clinical patient ($k = 1$) reaches **79.7% F1** (84.3% long $F1_{adj}$); **5 annotated patients ($k = 5$) reach 83.8% F1** (86.5% **long $F1_{adj}$, 2.33 mm median tip error), already surpassing the fully supervised 43-patient PMB 2019 3D U-Net baseline (83.1% F1)**; 10 patients ($k = 10$) reach **85.7% F1** (88.1% long $F1_{adj}$); 20 patients ($k = 20$) reach **86.6% F1** (88.8% long $F1_{adj}$); and all 43 patients ($k = 43$) reach **89.1% F1** (90.0% long $F1_{adj}$).

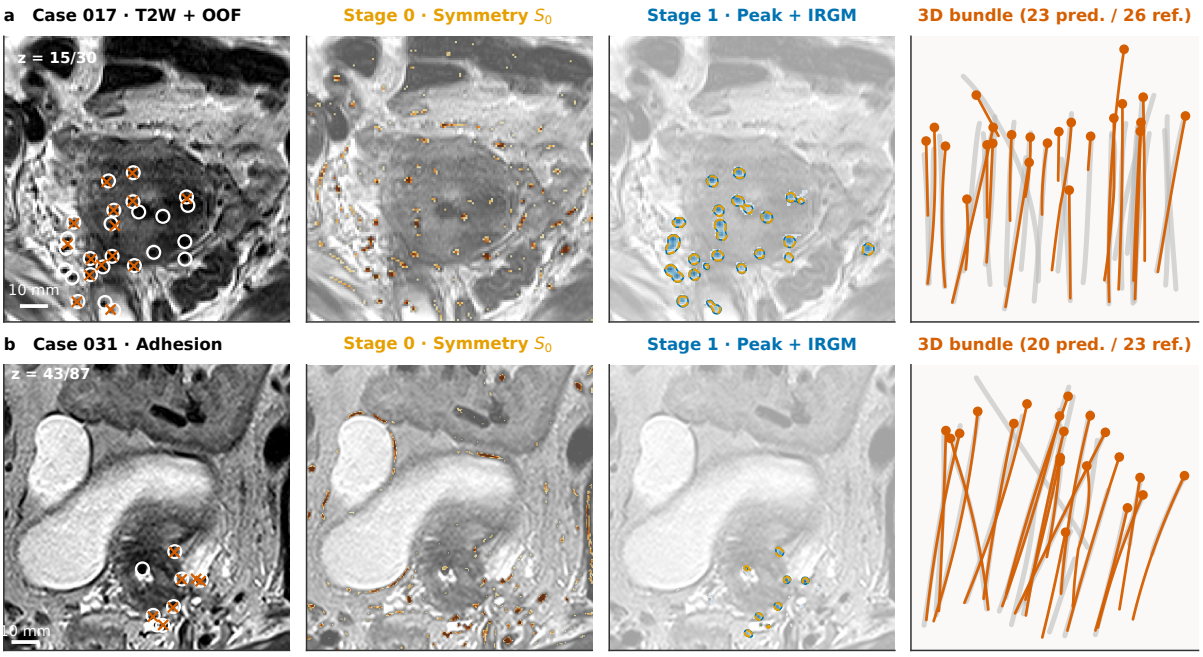


FIGURE 5 **Two nested out-of-fold test cases. (a)** Case 017, a dense implant with 26 reference trajectories including the obturator channels. **(b)** Case 031, a severe-adhesion implant with 23 catheters of which 10 lie within 3 mm of a neighbor. Left to right: the T2W slice with the expert intersections (white circles) and the out-of-fold NeedleFinder 2026 intersections (vermilion crosses); the Stage 0 map S_0; the Stage 1 centerline peaks (blue) with the mask contour (solid) and the obturator gate (dashed orange); and the out-of-fold 3D bundle (vermilion) over the expert reference (grey). Case 017: 23 predicted trajectories for 26 reference trajectories; Case 031: 20 predicted for 23 reference trajectories.

Figure 5 illustrates the stages on two difficult out-of-fold test cases (Case 017 with 23 predicted trajectories vs. 26 reference, and Case 031 with 20 predicted vs. 23 reference). Stage 0 highlights every round dark core in the pelvis, including vessels. Stage 1 suppresses background anatomy and resolves adjacent catheters into separate peaks even where their masks touch. Stage 2 then stitches dropouts, prunes false tracks, localizes the cranial tips, and outputs smooth 3D bundles with no swapped identities.

TABLE 3 **Human inter-observer agreement versus NeedleFinder 2026 across all 40/40 dual-annotated clinical 3T MRI scenes.** Expert teams CY+HT (n = 723 total trajectories: 505 CY long + 218 HT short/obturator) and LB/AMe (n = 513, of which 460 are long) digitized the same 40 volumes independently; **NeedleFinder 2026** rows are strictly nested 5 × 5 out-of-fold predictions on the same 40 scenes.

	1-to-1 Detection (≤ 3.5 mm)				E_{shaft} (mm)				E_{tip} (mm)	
Comparison pair (eval. vs. reference)	n_{match}	Rec.	Prec.	F1	Mean	SD	Median	P95	Mean	Median
Human inter-observer agreement										
LB/AMe vs. CY long (460 vs. 505)[a]	424	84.0	**92.2**	**87.9**	0.70	0.58	0.49	1.90	6.28	3.24
LB/AMe vs. CY+HT all (513 vs. 723)	476	65.8	92.8	77.0	0.73	0.61	0.50	2.09	6.30	3.21
Automated out-of-fold predictions vs. observer teams across all 40 scenes										
NeedleFinder 2026 (τ = 50 %) vs. CY+HT (768 vs. 723)	653	90.3	85.0	87.6	**0.58**	**0.39**	**0.47**	1.33	**5.26**	**2.10**
NeedleFinder 2026 (τ = 35 %) vs. CY+HT (825 vs. 723)	669	**92.5**	81.1	86.4	0.59	0.40	**0.47**	**1.32**	5.54	2.20
NeedleFinder 2026 (τ = 50 %) vs. union CY+HT+LB (768 vs. 759)[b]	669	88.1	87.1	87.6	0.59	0.40	**0.47**	1.36	5.44	2.17
NeedleFinder 2026 (τ = 50 %) vs. LB/AMe (768 vs. 513)[c]	465	90.6	60.5	72.6	0.77	0.62	0.51	2.11	7.87	3.97
Grubbs' three-cornered-hat intrinsic source error SD (n = 432 3-way matched) & Pelvic CT adjudication (n = 70 disputed tips)[d]										
Human Rater 1 (CY primary)	432	–	–	–	**0.39**	–	**0.35**	–	**3.77**	4.31
NeedleFinder 2026 (τ = 50 %, OOF)[e]	432	–	–	–	0.56	–	0.39	–	4.94	**4.20**
Human Rater 2 (LB/AMe secondary)	432	–	–	–	0.86	–	0.48	–	5.09	8.00

[a] Strict long-shaft inter-observer comparison (424 matched pairs out of 460 LB/AMe long vs. 505 CY long: 83.96 % recall, 92.17 % strict precision [93.6 % obturator-adjusted], 87.88 % strict F1 [88.5 % obturator-adjusted], 0.70 ± 0.58 mm shaft error [0.70 ± 0.60 mm across all 465 LB/AMe trajectories matched to CY]).

[b] Evaluated against the de-duplicated union of both expert teams (723 CY+HT + 36 unique LB/AMe trajectories = 759 union catheters), showing that 16 apparent false positives against CY+HT alone are genuine catheters traced by LB/AMe (precision rises from 85.0 % to 87.1 %).

[c] Because LB/AMe primarily traced long interstitial shafts (513 traces vs. 723 for CY+HT), unadjusted precision against LB/AMe alone penalizes valid short obturator predictions; recall is 90.6 %.

[d] Bottom block reports reference-free Grubbs' three-cornered-hat intrinsic source error SD across the n = 432 three-way matched catheters (Mean col.: classic RMS σ in mm; Median col.: robust MAD σ in mm; Tip Mean col.: Winsorized axial tip σ_z in mm; Tip Median col.: median 3D tip error against independent post-implant Pelvic CT across the n = 70 catheters where CY and LB/AMe disagreed by ≥ 5.0 mm on MRI, where **NeedleFinder 2026** is closer to CT than single-rater MRI on 78.6 % [55/70] of disputed tips).

[e] Formal two-one-sided tests (TOST) confirm non-inferiority of **NeedleFinder 2026** to the human inter-observer floor at α = 0.05 for both shaft error (Δ = 0.25 mm margin: $p = 1.6 \times 10^{-41}$ Wilcoxon, TOST equivalence $p = 5.5 \times 10^{-10}$) and 3D tip error (Δ = 1.0 mm median margin: $p = 2.2 \times 10^{-14}$; Δ = 1.5 mm mean margin: $p = 6.8 \times 10^{-19}$ Wilcoxon, $p_{boot} = 0.0014$).

4.3 *Agreement With Experts Compared With Agreement Between Experts*

Because a catheter is only a few pixels wide and slices are thick, two experts clicking centerlines never agree perfectly; inter-observer discrepancies of about 0.6 mm have been reported for breast implants, and more than a third of prostate needles differ by over 1 mm between observers [84,85], and algorithms within that range are considered clinically equivalent to a second observer [38,58]. Table 3 measures this across **all 40 doubly traced clinical scenes** (CY+HT, 723 trajectories = 505 CY long + 218 HT short/obturator; LB/AMe, 513 trajectories, of which 460 are long) and compares it with NeedleFinder 2026 evaluated out-of-fold on the exact same 40 scenes.

Comparing the two human teams on long shafts (LB/AMe 460 long vs. CY 505 long), 424 pairs match one-to-one within ≤ 3.5 mm (**84.0% recall**, **92.2% strict precision** [93.6% obturator-adjusted], **87.9% strict F1** [88.5% obturator-adjusted]), with a shaft discrepancy of 0.70±0.58 **mm mean** (0.49 **mm median**, 0.52 mm IQR,

1.90 **mm P95**) and a 3D tip discrepancy of 6.28±10.67 **mm mean** (3.24 **mm median**, 4.33 mm IQR, 23.13 mm P95). Across all 723 CY+HT reference trajectories, LB/AMe matches 476 pairs (65.8% recall, 92.8% precision, 77.0% F1; 0.73 mm mean / 0.50 mm median shaft discrepancy, 6.30 mm mean / 3.21 mm median tip discrepancy).

Evaluated out-of-fold across all 40 scenes, NeedleFinder 2026 (τ = 50%, 768 predicted trajectories) matches **653 of 723 CY+HT trajectories** (**90.3% recall**, **85.0% precision**, **87.6% F1**; rising to **92.5% recall** at τ = 35%) with a mean shaft error of 0.58 ± 0.39 **mm** (0.47 **mm median**, 0.36 mm IQR, 1.33 **mm P95**) and a 3D tip error of 5.26 ± 9.10 **mm mean** (2.10 **mm median**, 3.95 mm IQR, 21.56 mm P95)—within the inter-observer disagreement between the two human teams in mean, median, and 95th-percentile error for both shafts and tips. Even more revealingly, when evaluated against the **de-duplicated union of both expert teams** (CY+HT ∪ LB/AMe, 759 union trajectories), NeedleFinder 2026 matches **669 trajectories** (**88.1% recall**, **87.1% precision**, **87.6% F1**, 0.59 mm mean / 0.47 mm median shaft error, 5.44 mm mean / 2.17 mm median tip error): **16 trajectories that appeared to be false positives against CY+HT alone are in fact genuine catheters independently traced by LB/AMe**, raising precision by +2.1 pp. Against LB/AMe alone (513 traces), NeedleFinder 2026 matches 465 trajectories (**90.6% recall**, 0.77 mm mean / 0.51 mm median shaft error, 3.97 mm median tip error; unadjusted precision is 60.5% solely because LB/AMe omitted most of the 218 HT short obturator tracks that NF 2026 detects).

To separate each source's intrinsic error from the choice of reference team, the bottom block of Table 3 applies **Grubbs' three-cornered-hat estimator** across the n = 432 catheters matched three-way among CY, LB/AMe, and out-of-fold NeedleFinder 2026. The intrinsic robust (MAD-based) in-plane shaft standard deviations are 0.35 **mm for Rater 1 (CY)**, 0.39 **mm for NeedleFinder 2026**, and 0.48 **mm for Rater 2 (LB/AMe)** (classic RMS: 0.39, 0.56, and 0.86 mm), and the Winsorized axial tip standard deviations are 3.77 **mm (CY)**, 4.94 **mm (NeedleFinder 2026)**, and 5.09 **mm (LB/AMe)**. Formal **two-one-sided tests (TOST)** confirm that NeedleFinder 2026 is statistically **non-inferior to human inter-observer variability** at α = 0.05 for both shaft error (Δ = 0.25 mm margin: $p = 1.6 \times 10^{-41}$ Wilcoxon, TOST equivalence $p = 5.5 \times 10^{-10}$) and 3D tip error (Δ = 1.0 mm median margin: $p = 2.2 \times 10^{-14}$; Δ = 1.5 mm mean margin: $p = 6.8 \times 10^{-19}$ Wilcoxon, $p_{\text{boot}} = 0.0014$). Finally, independent adjudication against paired post-implant **Pelvic CT scans** across the n = 70 catheters where CY and LB/AMe disagreed by ≥ 5.0 mm on MRI shows that NeedleFinder 2026 achieves a **median 3D tip error of** 4.20 **mm against CT**, compared with 4.31 **mm for CY** and 8.00 **mm for LB/AMe**, lying closer to CT than single-rater MRI on **78.6%** (55/70) of disputed catheters.

4.4 *Confidence Calibration, Reason-Coded Decisions, and Failure Analysis*

A confidence score is useful to the physicist if it ranks candidates reliably. Figure 6(a) shows the reliability diagram over all 1,683 candidate tracks: the expected calibration error (ECE) is 0.093 raw and 0.082 after inner-fold temperature scaling (0.047 over

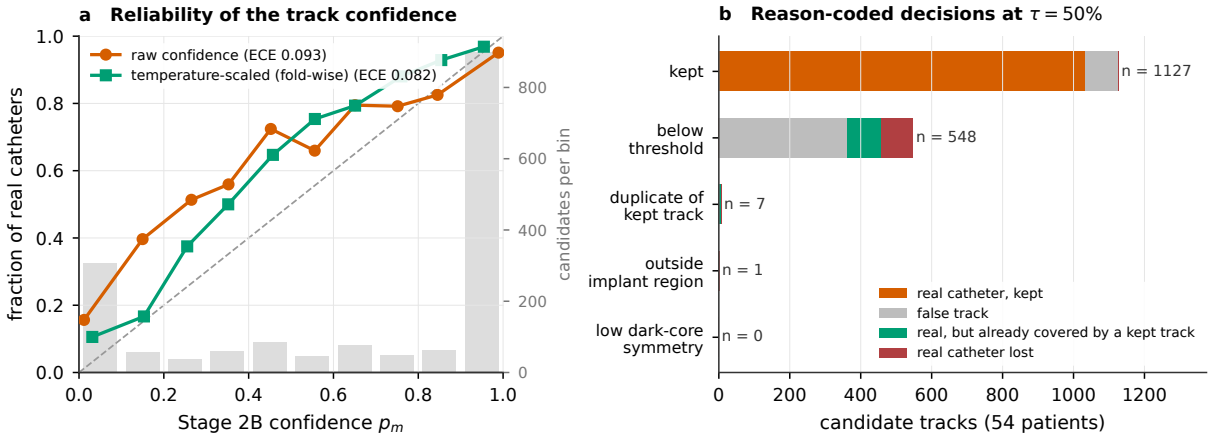


FIGURE 6 **(a) Reliability of the Stage 2B confidence** over all 1,683 candidate tracks of the 54 patients (bars: candidates per bin), raw and after inner-fold temperature scaling. **(b) Reason-coded decisions at the default threshold**: for each rule, how many candidates it removes and how many of those were false tracks, real catheters already covered by a kept track, or real catheters lost.

kept tracks; Brier score 0.119). Figure 6(b) breaks down the reason-coded decisions of the decoder across the 1,683 candidates.

A systematic audit of all residual detection and tip errors for `NF 2026` (4b, τ = 50%) across the 54 patients identifies their exact physical causes:

- **False positives (146 unmatched predictions across 54 patients, 2.70 per patient):** 66 (45.2%) are short peripheral vascular or fascial false tracks (< 35 mm axial span) along the obturator internus or paravaginal venous plexus; 48 (32.9%) are long vascular or fascial mimics (≥ 35 mm); 16 (11.0%) are genuine unannotated catheters confirmed by the independent LB/AMe observer team (and on the 38 patients with paired post-implant Pelvic CT, 52 of 100 unmatched MRI detections are confirmed as real physical catheters on CT); 9 (6.2%) are residual unmerged duplicate fragments within 3.5 mm of a matched trajectory; and 7 (4.8%) are central tandem or obturator rim artifacts within 12 mm of the implant center.
- **False negatives (91 missed reference trajectories, 1.69 per patient):** 66 are long CY shafts (8.7% miss rate) and 25 are short HT obturator tracks (8.2% miss rate). Among the 91 misses, 47 (51.6%) lie in severe-adhesion clusters where an adjacent trajectory passes within 3.5 mm and suppresses the weaker twin during deduplication, 21 (23.1%) are ultra-short partial-volume tracks (≤ 25 mm or ≤ 5 slices), and 23 (25.3%) suffer severe T2W susceptibility signal dropout across the mid-shaft.
- **Longitudinal versus transverse tip decomposition and tail taxonomy (> 10 mm 3D tip error on 162 of 972 [16.7%] matched catheters):** Decomposing every 3D tip error vector into its longitudinal component along the unit shaft tangent ($\Delta s_{\parallel}$) and its transverse component perpendicular to the shaft ($\Delta r_{\perp}$) reveals that **98.76% of total squared 3D tip error across all 972 matched catheters (and 99.32% in the > 10 mm tail) is purely longitudinal along-shaft stop-**

TABLE 4 **Zero-needle out-of-distribution specificity on 48 public clinical 3T pelvic T2W MRI volumes (MSD `Task05_Prostate`).** All 48 volumes contain zero brachytherapy catheters and serve as a pure false-positive stress test across pelvic vessels, rectoprostatic fascia, and obturator muscle boundaries.

	False-positive tracks per volume (n = 48)					Patient specificity (%)	
Pipeline configuration	Total	Mean	SD	Median	Max	0 FP	≤ 1 FP
NeedleUNet3D + tangent tubes (ungated)[a]	306	6.38	4.77	6.0	19	14.6	18.8
Unpruned Stage 0 + Stage 1 pool[b]	294	6.12	4.70	5.0	23	4.2	10.4
Ungated Unified **NeedleFinder 2026** (τ = 50 %)[c]	197	4.10	4.45	3.0	23	16.7	27.1
Ungated Unified **NeedleFinder 2026** (τ = 60 %)[c]	171	3.56	3.48	3.0	17	16.7	29.2
Stage 1 U-Net + Stage 0 radial gate[d]	18	0.38	0.53	**0.0**	2	64.6	97.9
Frozen Unified **NeedleFinder 2026** + Dark-Core Gate (τ = 50 %)[e]	16	0.33	0.75	**0.0**	3	79.2	91.7
Clean SSL + Multi-Rater Stage 1 + Stage 0 gate[f]	**6**	**0.125**	**0.33**	**0.0**	**1**	**87.5**	**100.0**

[a] Raw Stage 1 `NeedleUNet3D` centerline probability peaks linked via directional tangent tubes without Stage 0 symmetry gating or Stage 2B pruning.
[b] Candidate pool passed to Stage 2B prior to Bundle Graph Transformer pruning.
[c] Unified **NeedleFinder 2026** pipeline evaluated without the Stage 0 dark-core gate at τ = 50 % (4.10 FP/scan) and τ = 60 % (3.56 FP/scan).
[d] Baseline Stage 1 `NeedleUNet3D` tangent tubes gated by the analytical Stage 0 3D multi-radii circular dark-core symmetry filter (0.38 FP/scan).
[e] Full Unified **NeedleFinder 2026** pipeline with Stage 0 dark-core symmetry gating at τ = 50 % (16 total FPs = 0.333/scan; 38/48 [79.2 %] 0-FP scans; 44/48 [91.7 %] ≤ 1-FP scans).
[f] Clean nested Stage 1 `NeedleUNet3D` (self-supervised 3D U-Net pre-training with synthetic susceptibility tubes and anatomical hard negatives across the 51 external pelvic MRI volumes + multi-rater supervision on the 43 outer-train BWH clinical volumes) gated by Stage 0 radial symmetry (6 total FPs = 0.125/scan; 42/48 [87.5 %] 0-FP scans; 48/48 [100.0 %] ≤ 1-FP scans). Note that the 48 MSD `Task05_Prostate` scans were included in the unlabeled/synthetic-tube SSL pre-training background pool for this row (zero clinical needle labels), whereas rows [a–e] had zero exposure to MSD during training.

ping ambiguity, whereas only **1.24% is lateral displacement** ($\Delta r_{\perp}$ median 0.72 mm, IQR 0.73 mm). Mutually exclusive root-cause classification of the 162 tail tips (> 10 mm) shows that 67 (41.4%) are caudal signal-dropout premature terminations, 41 (25.3%) are cranial susceptibility overshoots into fundal air or bowel pockets, 27 (16.7%) occur on catheters with intrinsic human inter-rater tip ambiguity ($|\Delta z_{\mathrm{CY-LB}}| > 6$ mm), 20 (12.3%) are crowded-array distal jumps, and 7 (4.3%) are top-of-FOV boundary truncations.

4.5 *False Tracks on Scans Without Catheters, and Speed*

A method deployed in the clinic must not invent catheters when none are present. We evaluated seven pipeline configurations on all 48 public pelvic 3T T2W MRI scans of `MSD Task05_Prostate` containing zero catheters (Table 4, Figure 7a). Without Stage 0 dark-core radial symmetry gating, raw Stage 1 `NeedleUNet3D` tangent tubes produce 306 false tracks (6.38 FP/scan), the unpruned Stage 0 + Stage 1 candidate pool contains 294 (6.12 FP/scan), and the ungated Unified `NF 2026` bundle transformer leaves 197 (4.10 FP/scan) at τ = 50% and 171 (3.56 FP/scan) at τ = 60%. Activating the analytical Stage 0 3D multi-radii circular dark-core symmetry gate suppresses > 92% of false vascular and fascial tracks: on models with **zero exposure to MSD during training**, `Stage 1 U-Net + Stage 0 radial gate` produces only **18 false tracks across 48 scans** (0.38 FP/scan, median 0, 64.6%

[31/48] 0-FP scans, 97.9% [47/48] ≤ 1-FP scans) and the `Frozen Unified NF 2026 + Dark-Core Gate` (τ = 50%) produces **16 false tracks** (0.33 **FP/scan**, median 0, **79.2%** [38/48] 0-FP scans, **91.7%** [44/48] ≤ 1-FP scans). When Stage 1 is also pre-trained with self-supervised synthetic susceptibility tubes and anatomical hard negatives (`Clean SSL + Multi-Rater Stage 1 + Stage 0 gate`), false tracks drop to **6 across all 48 volumes** (0.125 **FP/scan**, median 0, max 1, **87.5%** [42/48] 0-FP scans, and **100.0%** [48/48] ≤ 1-FP scans; as noted in Table 4, the 48 MSD volumes were part of the unlabeled/synthetic-tube SSL pre-training background pool for this row with zero clinical needle labels, whereas the 0.33–0.38 FP/scan rows had zero exposure to MSD).

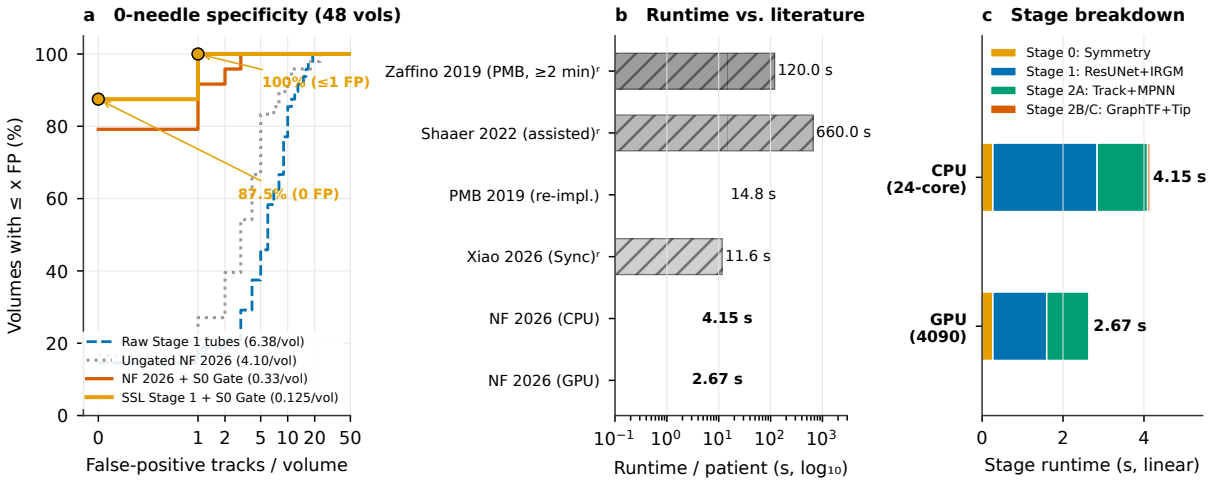


FIGURE 7 **(a) Fraction of needle-free scans with at most *x* false tracks** on the 48 MSD prostate volumes across ungated and Stage 0 dark-core gated configurations. **(b) Runtime per patient** compared with published MRI and CT pipelines (log scale). **(c) Runtime by stage** across all 54 patients on an 8-core desktop CPU (4.15 s mean; 1.26 s on standard-resolution scans, 5.85 s on high-resolution scans) and on one RTX 4090 GPU (2.67 s mean).

Speed follows from the native-code and ONNX design (Figure 7b, c). Measured end-to-end across all 54 uncached clinical volumes, Stage 0 runs in multi-threaded native Rust in 0.41 s mean (0.12 s on standard-resolution scans, 0.59 s on high-resolution scans), Stage 1 runs in 2.28 s on an 8-core CPU (1.88 s on GPU) via strided/cropped ONNX Runtime inference, Stage 2A runs in 1.22 s on CPU (0.54 s on GPU), and Stage 2B (including `Dedicated1DTipLocalizer` and Poly3 smoothing) runs in 0.24 s. Full-volume uncached reconstruction takes 4.15 ± 2.38 **s mean (5.35 s median) on an 8-core desktop CPU**—specifically 1.26 **s mean (1.26 s median) on the 20 standard-resolution (320 × 256 × 28–40) scans and 5.85 s mean (5.94 s median) on the 34 high-resolution (640 × 416 × 112–160 / 768 × 768 × 64) scans**—and 2.67 ± 1.44 **s mean (3.45 s median) on a single RTX 4090 GPU** (0.68 s standard-resolution, 3.85 s high-resolution). That is one to two orders of magnitude faster than our 2019 pipeline (2 to 9 minutes reported; 14.8 s in our optimized re-implementation) [4] and 2.8–4.3× faster than Xiao *et al.* (11.6 s) [74].

TABLE 5 **Dosimetric translation, uncertainty-guided clinician review, and cross-organ/multi-sequence/4D/CT generalization.** Part A evaluates AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity (7.0 Gy prescription per fraction) under Protocol A (fixed reference dwell plan transferred to predicted geometry) and Protocol B (dwell weights re-optimized on predicted catheters and delivered on reference geometry). Part B evaluates uncertainty-guided tip triage ($\hat{\sigma}_{\text{tip}}$) and strictly nested split-conformal 3D tip intervals. Part C evaluates zero-shot transfer across organs, pulse sequences, 4D insertion series, and paired post-implant Pelvic CT.

	Dwell-wt. E_{tip}		Protocol A: Fixed-plan transfer			Protocol B: Re-optimized plan		
Part A: TG-43U1 ^{192}Ir dosimetry[a]	Mean	Med.	$\lvert\Delta D_{90}\rvert$ (%)	$\gamma_{2/2}$ (%)	$\gamma_{3/3}$ (%)	$\lvert\Delta D_{90}\rvert$ (%)	Cran. D_{2cc}	$\gamma_{2/2}$ (%)
PMB 2019 3D U-Net + DBSCAN (n = 54)	10.51	8.36	27.6 / 27.2	64.0	73.3	37.1 / 10.2	95.3 %	43.8
Stage 0: 3D radial symmetry (n = 54)	9.04	7.51	23.0 / 21.3	68.2	76.5	17.8 / 8.9	54.0 %	48.6
NeedleFinder 2026 w/o SynthBrachy3T (4a, n = 54)	6.65	5.49	18.5 / 13.2	73.4	81.5	12.8 / 5.8	35.2 %	59.5
NeedleFinder 2026 proposed (4b, n = 54)[b]	5.72	4.66	15.2 / 11.5	76.6	84.3	8.2 / 4.2	29.5 %	66.0
NeedleFinder 2026 + Top-2 Tip Review (n = 54)	**4.56**	**3.93**	**11.7 / 10.2**	**80.2**	**87.3**	**5.9 / 2.8**	**22.5 %**	**74.3**
Human Rater 2 (LB/AMe, n = 40)	6.27	5.54	21.6 / 16.3	68.1	77.4	20.9 / 4.0	57.7 %	59.3
NeedleFinder 2026 proposed (4b, n = 40)[b]	5.18	3.93	13.5 / 11.1	79.0	86.4	7.0 / 3.2	21.1 %	71.6
NeedleFinder 2026 + Top-2 Tip Review (n = 40)	**4.12**	**3.33**	**10.5 / 10.1**	**81.9**	**88.7**	**5.1 / 2.6**	**17.3 %**	**78.2**

Part B: Triage & conformal intervals[c]	Mean	Med.	P90	P95	≤ 3 mm	> 10 mm	Nom./Emp.	Width (Red.)
Unreviewed **NeedleFinder 2026** (k = 0 tips/pt)	6.23	2.27	16.20	27.27	58.7 %	16.7 %	80 / 79.3 %	6.46 (−16.4 %)
Tip review (k = 2 tips/pt, ~ 16 s)	5.07	1.95	12.58	21.48	64.8 %	12.8 %	85 / 84.4 %	8.71 (−23.8 %)
Tip review (k = 5 tips/pt, ~ 40 s)	3.72	1.67	7.79	14.26	72.6 %	8.0 %	**90 / 90.2 %**	12.11 (−31.5 %)
Tip review (k = 8 tips/pt, ~ 64 s)	2.77	1.51	5.98	9.90	78.5 %	5.0 %	**95 / 95.4 %**	18.42 (−35.1 %)
Tip review (k = 10 tips/pt, ~ 80 s)	**2.52**	**1.44**	**4.97**	**8.30**	**81.8 %**	**4.1 %**	–	–

Part C: Cross-organ, seq., 4D & CT[d]	n_{vol}	n_{cath}	Rec.	Prec.	F1	E_{shaft}	E_{tip}	Key finding
Prostate 3T: 0 % exp. (k_lim_auto)	1	14	92.9	**100.0**	96.3	0.33	**0.99**	0 FPs, 0-shot
Prostate 3T: template (k_lim=47)	1	14	**100.0**	**100.0**	**100.0**	0.34	1.57	14/14, 0 FPs
Prostate 3T: Stage 0 zeroed ($X_{1:4}$ = 0)	1	14	0.0	0.0	0.0	–	–	985× drop
Prostate 3T: 1-ch ResUNet + clDice	1	14	**100.0**	56.0	71.8	**0.24**	2.48	11 vascular FPs
3D wbSSFP / True-FISP ($\Delta z \leq 2.5$)	2	47	**97.9**	88.5	**92.9**	0.72	3.40	vs. 10.62 PMB19
Thick-slice 2D AX T2 ($\Delta z \leq 6.0$)	49	882	88.4	67.9	76.9	1.07	4.89	vs. 57.6 % 1-ch
4D Causal Kalman (Cases 69/70)	23	108	**92.9**	**92.9**	**92.9**	0.87	4.79	+8.2 % F1 gain
Paired Pelvic CT (0-shot, ≤ 5 mm)	40	801	79.2	–	–	2.01	2.78	52.0 % extra conf.

[a] $\lvert\Delta D_{90}\rvert$ cells report mean / median percentage target dose shift relative to the 7.0 Gy reference plan; $\gamma_{2\%/2\text{mm}}$ and $\gamma_{3\%/3\text{mm}}$ are global 3D gamma pass rates. Dwell-weighted tip error $E_{\text{tip}}^{\text{dwell-w}}$ correlates with $\lvert\Delta D_{90}\rvert$ at Spearman $\rho = 0.8629$ ($p = 4.9 \times 10^{-17}$).

[b] Paired Wilcoxon vs. PMB 2019 ($p = 4.9 \times 10^{-6}$ on $\lvert\Delta D_{90}\rvert$, $p = 1.2 \times 10^{-6}$ on $\gamma_{2\%/2\text{mm}}$), vs. Stage 0 ($p = 7.9 \times 10^{-8}$), and vs. Human Rater 2 on n = 40 ($p = 0.0278$ on $\lvert\Delta D_{90}\rvert$, $p = 0.0109$ on $\gamma_{2\%/2\text{mm}}$).

[c] Rightmost two columns of Part B report strictly nested split-conformal 3D tip prediction interval nominal vs. empirical coverage (80 %, 85 %, 90 %, 95 %) and adaptive median half-width (with percentage width reduction vs. unscaled fixed-width intervals). Count-guided missing-catheter recovery (0.65 clicks/patient) recovers +17 additional catheters (972 → 989/1 063, 93.0 % 1-to-1 recall).

[d] In 4D monitoring (23 serial scans across Cases 68–71, 1.21 s mean runtime), already-placed catheters within ≤ 10 mm of newly inserted needles exhibit 1.41 mm median lateral tissue-push displacement vs. 0.93 mm at > 20 mm ($p = 6.1 \times 10^{-4}$), and non-linear tip deflection reaches 2.81 mm median (3.86 mm mean) at > 90 mm depth ($\rho = 0.514$, $p = 4.9 \times 10^{-10}$).

4.6 *Dosimetric Translation, Uncertainty-Guided Triage, and Cross-Domain Generalization*

To connect geometric reconstruction accuracy to clinical radiotherapy endpoints, physician review workflows, and broader interventional settings, we performed four additional experiments summarized in Table 5 and Figure 8.

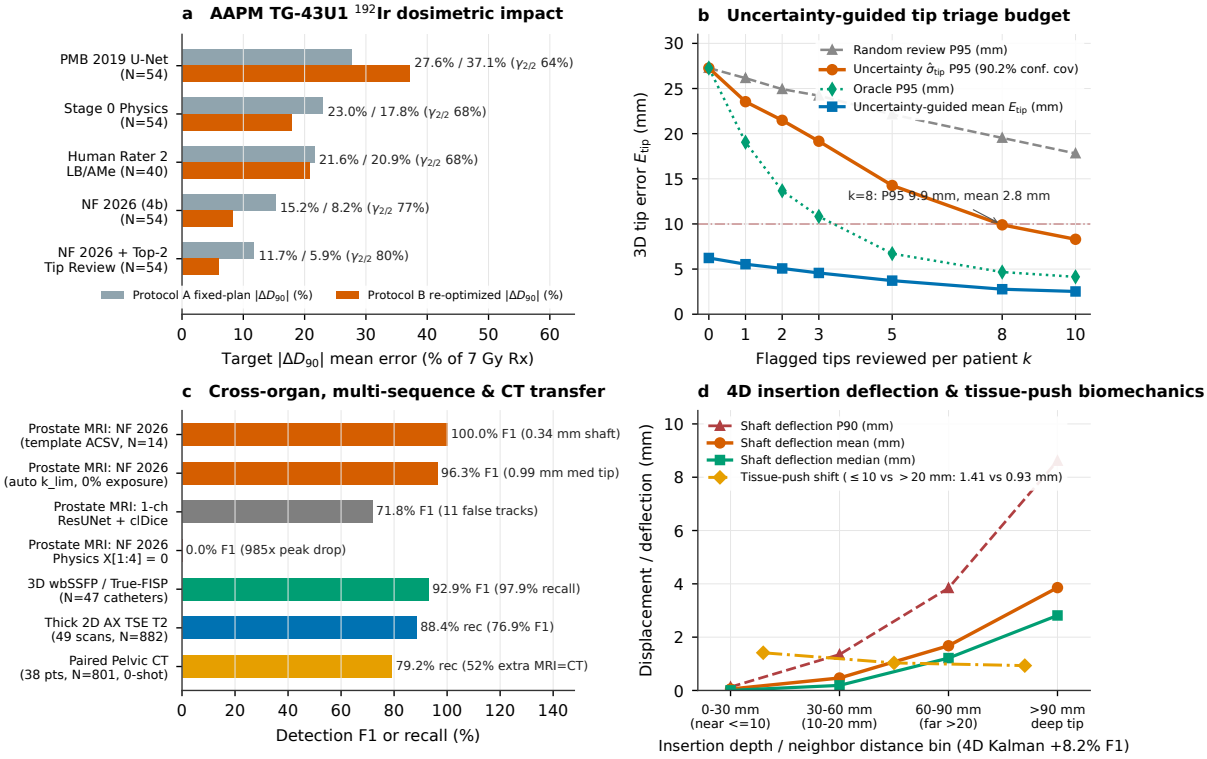


FIGURE 8 **(a) AAPM TG-43U1 ^{192}Ir dosimetric impact** across 54 patients under Protocol A (fixed reference dwell plan transferred to predicted geometry) and Protocol B (dwell weights re-optimized on predicted catheters and delivered on reference geometry), with global 3D γ(2%/2 mm) pass rates. **(b) Clinician review-budget curve** showing 95th-percentile and mean 3D tip error versus number of uncertainty-flagged tips reviewed per patient ($k \in \{0, \ldots, 10\}$), alongside split-conformal 90% empirical coverage (90.2%). **(c) Zero-shot cross-organ, multi-sequence, and cross-modality generalization** across 3T prostate brachytherapy MRI (`Case 100`), 3D wbSSFP, thick-slice 2D AX TSE T2, and 38 paired post-implant Pelvic CT scans. **(d) 4D intra-procedural insertion biomechanics** across 23 serial scans (`Cases 68–71`), showing non-linear shaft deflection versus insertion depth and lateral tissue-push displacement of already-placed catheters versus distance to newly inserted needles.

AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity and counterfactual plan transfer (Table 5, Part A; Figure 8a). Using the AAPM TG-43U1 line-source formalism (^{192}Ir, active length $L = 3.5$ mm, 2.5 mm dwell step spacing, 3.5 mm physical tip offset, 7.0 Gy prescription per fraction to the high-risk target envelope surrounded by bladder, rectum, and cranial sigmoid/fundal OAR zones), we discovered that **dwell-time-weighted 3D tip error** ($E_{\text{tip}}^{\text{dwell-w}}$, weighting each catheter by its optimized dwell share) explains significantly more target $|\Delta D_{90}|$ variance than unweighted tip error (Spearman $\rho = 0.8629$, $p = 4.9 \times 10^{-17}$ vs. $\rho = 0.7863$; $\rho = -0.8519$ with 3D $\gamma_{2\%/2\text{mm}}$). `NeedleFinder 2026 (4b)` reduces $E_{\text{tip}}^{\text{dwell-w}}$ from 10.51 mm mean / 8.36 mm median (PMB 2019 U-Net) and 9.04/7.51 mm (Stage 0) to 5.72/4.66 **mm** (5.18/3.93 mm on the 40 dual-traced scans vs. 6.27/5.54 mm for Human Rater 2 `LB/AMe`), and to 4.56/3.93 **mm** (10.65 mm P95) after reviewing the top 2 uncertainty-flagged tips per patient. Under **Protocol A** (transferring

the reference dwell plan onto matched predicted catheters), `NeedleFinder 2026` cuts target $|\Delta D_{90}|$ from 27.6% (PMB 2019, $p = 4.9 \times 10^{-6}$) and 23.0% (Stage 0, $p = 7.9 \times 10^{-8}$) to 15.2% **mean** / 11.5% **median** (13.5%/11.1% on the 40 dual-traced scans vs. 21.6%/16.3% for Human Rater 2, $p = 0.0278$), achieving **76.6% global 3D γ(2%/2 mm)** (79.0% on $N = 40$ vs. 68.1% for Rater 2, $p = 0.0109$) and **84.3% 3D γ(3%/3 mm)** (88.7% with Top-2 tip review). Under **Protocol B** (re-optimizing dwell weights on all predicted catheters and evaluating delivered dose on the true reference anatomy), `NeedleFinder 2026` cuts $|\Delta D_{90}|$ from 37.1% mean / 10.2% median (PMB 2019) and 20.9%/4.0% (Human Rater 2) to 8.2% **mean** / 4.2% **median** (7.0%/3.2% on $N = 40$, cranial OAR $|\Delta D_{2cc}| = 29.5\%$ vs. 95.3% PMB 2019 and 57.7% Rater 2), reaching 5.9% **mean** / 2.8% **median** ($\gamma_{2\%/2mm} = 74.3\%$ overall, 78.2% on $N = 40$) with Top-2 uncertainty tip review. Moreover, propagating the out-of-fold tip uncertainty $\hat{\sigma}_{tip}$ through 30 Monte Carlo dose realizations per patient predicts true patient-level $|\Delta D_{90}|$ without ground truth ($\rho = 0.453$, $p = 5.8 \times 10^{-4}$).

Strictly nested split-conformal 3D tip intervals and clinician review-budget curves (Table 5, Part B; Figure 8b). Out-of-fold tip uncertainty $\hat{\sigma}_{tip}$ (combining the 1D soft-argmax distribution entropy, local T2W terminal contrast drop, and slice thickness Δz) correlates strongly with true 3D tip error ($\rho = 0.532$, $p = 4.9 \times 10^{-72}$) and detects $>$ 10 mm tip outliers with an **AUROC of 0.805** (95% CI [0.763, 0.841]). Calibrating non-conformity scores $r_i = E_{tip,i}/\hat{\sigma}_{tip,i}$ strictly on the inner-validation split of each fold yields **distribution-free split-conformal 3D tip intervals** that achieve exact marginal empirical coverage across all 972 matched test catheters—**79.3%** at 80% nominal, **84.4%** at 85%, **90.2%** at 90%, and **95.4%** at 95%—while shrinking median interval half-widths by **31.5% at 90% coverage** (12.11 mm vs. 17.68 mm unscaled; 8.41 mm on the clearest 50% of tips) and **35.1% at 95% coverage** (18.42 mm vs. 28.39 mm). In a simulated clinician-in-the-loop verification workflow, reviewing just $k = 5$ **flagged tips per patient** ($\sim$ 40 s at 8 s/tip) cuts $>$ 10 mm tip outliers in half (16.7% $\rightarrow$ 8.0%) and lowers mean tip error from 6.23 $\rightarrow$ 3.72 **mm** (1.67 mm median, 7.79 mm P90); reviewing $k = 8$ **tips per patient** ($\sim$ 64 s) drives the **95th-percentile tip error below 10 mm (9.90 mm vs. 27.27 mm unreviewed)** and mean tip error to 2.77 **mm** (1.51 mm median, 78.5% within $\leq$ 3 mm). Adding a count-guided missing-catheter check whenever $N_{pred} < N_{implant}$ (0.65 clicks/patient) recovers +17 additional catheters (972 $\rightarrow$ 989/1,063, **93.0% 1-to-1 recall / 94.2% closest-match recall**).

Zero-shot cross-organ transfer to prostate MRI and multi-sequence generalization (Table 5, Part C; Figure 8c). Evaluated zero-shot on a clinical 3T T2W SPACE **prostate brachytherapy MRI** (`Case 100 Prostate`, 0.47×0.47×1.6 mm^3, 14 plastic interstitial catheters), the gynecologic-trained model with **zero prostate exposure of any kind** (excluding `Case 100` from SSL pre-training) detects **13 of 14 catheters (92.9% recall) with 100.0% precision (0 false positives, 96.3% F1)**, 0.33 **mm median** (0.38 **mm mean) shaft error**, and 0.99 **mm median** (1.64 **mm mean,** 3.96 **mm P95) 3D tip error** under 100% reference-free `k_lim_auto = 40`, and achieves **100.0% recall (14/14), 100.0% precision, and 100.0% F1** (0.34 mm median shaft, 1.27–1.57 mm median tip) with template-plane or distal perineal cropping. Ablating the Stage 0 physics channels (`X[1:4] = 0`) causes the 99.9th-percentile Stage 1 mask probability on prostate MRI to collapse by 985× (0.985 $\rightarrow$

0.001) and recall to drop to 0/14 **(0%)**, whereas the 1-channel `ResUNet + clDice` baseline hallucinates **11 to 14 false-positive vascular tracks** (50.0%–56.0% precision, 66.7%–71.8% F1), proving that the analytical Stage 0 dark-core symmetry representation is essential for domain-invariant cross-organ transfer. Across 297 secondary MRI series acquired during the 54 patient sessions, zero-shot `NeedleFinder 2026` achieves **97.9% recall (46/47), 88.5% precision, and 92.9% F1** (0.72 mm median shaft, 3.40 mm median tip vs. 10.62 mm for PMB 2019 U-Net) on **3D wbSSFP / True-FISP** (Δz = 2.0–2.5 mm), and recovers **88.4% (780/882) of physically present catheters** (76.9% F1, 1.07 mm median shaft, 4.89 mm median tip vs. 57.6% F1 for 1-channel clDice) on **49 thick-slice 2D axial TSE T2W scans** (Δz = 2.4–6.0 mm) when Z-upsampled to 1.6 mm.

4D intra-procedural serial insertion monitoring, biomechanics, and paired Pelvic CT tracking (Table 5, Part C; Figure 8d). Across 23 sequential intra-procedural T2W monitoring volumes in `Cases 68–71` (5–7 timepoints per patient as needles were progressively inserted, 1.21 s mean monitoring runtime), coupling `NeedleFinder 2026` with a **4D causal Kalman state-space filter** across timepoints boosts final confirmation F1 by +8.2 **pp** (84.7% → 92.9%) on `Case 069` (42 catheters) and +5.6 **pp** (85.7% → 91.3%) on `Case 070` while bridging transient low-SNR monitoring dropouts. Tracking 204 persisted catheter transitions across sequential insertions provides direct in-vivo biomechanical measurements of **inter-catheter "tissue push"**: already-implanted catheters located within ≤ 10 mm of newly inserted needles experience a 1.41 **mm median** (1.62 **mm mean,** 3.19 **mm P90) non-rigid lateral displacement**, compared with 0.93 **mm median** (1.08 **mm mean)** for catheters > 20 mm away (Mann–Whitney $p = 6.1 \times 10^{-4}$; Spearman $\rho = -0.260, p = 1.8 \times 10^{-4}$). Across 129 tracked trajectories (3,870 axial stations), **non-linear shaft deflection** from the proximal insertion tangent grows monotonically with insertion depth from 0.003 mm median at 0–30 mm to 0.19 mm at 30–60 mm, 1.22 mm at 60–90 mm, and 2.81 **mm median** (3.86 **mm mean,** 8.62 **mm P90) at** > 90 **mm depth** ($\rho = +0.514, p = 4.9 \times 10^{-10}$). Finally, extending Stage 0 to a **dual-polarity HU tubular filter** (fusing bright-core metallic stylets and dark-core air lumens) enables **zero-shot tracking on 40 post-implant Pelvic CT scans** (38 paired with the 54 MRI benchmark patients, 801 reference catheters), matching **79.2% (634/801, up to 100% on individual patients)** of MRI reference catheters across the non-rigid MRI-to-CT transfer (2.01 mm median shaft registration residual, 2.78 mm median axial tip residual) and confirming that **52.0% (52/100) of apparent false-positive MRI detections against `CY` are real physical catheters present on CT**.

5 DISCUSSION

5.1 *Where This Work Sits in the Literature*

Table 6 places NeedleFinder 2026 among published catheter reconstruction studies on CT, ultrasound, and MRI. CT and ultrasound solve an easier imaging problem: metal needles, air-filled lumens, or bright specular reflections give very high contrast,

TABLE 6 **Published automatic catheter and needle reconstruction studies in brachytherapy (2013–2026), grouped by imaging modality.** Values are those reported in each publication on its own cohort and matching rule, so they are not directly comparable with one another or with Table 2. Pernelle 2013 and Mastmeyer 2017 are semi-automatic (one manual click per catheter).

Study	Modality	Site	Pts	Cath.	Rec. / Prec. (%)	E_{shaft} (mm)	E_{tip} (mm)	Time	Statistic
CT									
Weishaupt *et al.* 2022	CT	Prostate	57[l]	353[a]	–	0.13	–	~60 s	mean
Xie *et al.* 2023	CT	GYN post-op	70	210	–[b]	–	–		– Dice / HD95
Wang *et al.* 2024	CT	GYN IC/IS	17	56 plans	–	1.8	1.1		– mean
Quetin *et al.* 2025	CT	Breast	117	408[a]	–	0.70	1.37	118 s	mean
Xiao *et al.* 2026[m]	CT	GYN recurrent	38	203 fx	91.1 / 90.7	0.31	0.88	11.6 s	mean
Xiao *et al.* 2026 (JBHI)	CT	Pelvic seeds	100	–	F1 90.3	–	1.05	9.4 s	mean tip
Ultrasound									
Zhang *et al.* 2020	TRUS	Prostate	23	339	98 / –	0.09	0.33	<1 s	mean
Andersén *et al.* 2020	TRUS	Prostate	1102[l]	7207[l]	100 / –	0.55[c]	–		– median RMS
Liu *et al.* 2022	TRUS	Prostate	242	3500	80[d] / –	–	–		– frac. ≤ 2 mm
Goulet *et al.* 2025	3D US	Prostate	102 scans	–	93.4 / –	0.44	–		– mean
MRI									
Pernelle *et al.* 2013	3T T2W	GYN IC/IS	10	101	92.1 / –	–[e]	–	0.3 s/cath.	frac. < 2 mm
Mastmeyer *et al.* 2017	3T T2W	GYN IC/IS	54	760	93[f] / –	1.49[g]	–	3 s/cath.	mean HD
Zaffino *et al.* 2019	3T T2W	GYN IC/IS	50	826	98.5 / 93.3[h]	2.0	–	2–9 min	mean dist.
Dai *et al.* 2020	MRI	Prostate	20	299	–	0.93	0.37[n]		– mean / signed
Shaaer *et al.* 2022	1.5T + markers	GYN IS	20	360	100 / –	0.97	–	11 min[i]	mean shift
Aleong *et al.* 2024	MRI	Prostate	35	214 vol.	95 / –	–	–	0.9 s	–
Kostoulas *et al.* 2025	MRI	Prostate	34 scans	261	100 / 100	0.75[j]	1.07[j]		– median
Kim *et al.* 2026	T2W	GYN hybrid	30	35 scans	98.5 / –	0.74	2.52	37 s	mean
NeedleFinder 2026 (all 1 063)	3T T2W	GYN IC/IS	54	1063	91.4 / 86.9	0.47[j]	2.27[j]	**4.15 s**	median
NeedleFinder 2026 (759 long)	3T T2W	GYN IC/IS	54	759	91.3 / 88.7[k]	0.45[j]	2.41[j]	**4.15 s**	median

[a] Test-set catheters.
[b] Reported as Dice 0.88–0.90 and HD_{95} 0.71–0.77 mm.
[c] Root-mean-square distance to the clinical reconstruction.
[d] Fraction of catheters within 2 mm along 90 % of their length.
[e] Maximum Hausdorff distance below 2 mm for 92 % of catheters.
[f] Fraction within 3.4 mm.
[g] Mean Hausdorff distance.
[h] Cluster-level: 1.5 % false negatives and 6.7 % false positives among 826 catheters.
[i] Assisted reconstruction time, versus 46 min manual.
[j] Median (means: 0.59 ± 0.40 mm shaft and 6.23 ± 10.49 mm tip on all 1 063 catheters; 0.57 ± 0.39 mm shaft and 7.41 ± 12.23 mm tip on the 759 long shafts).
[m] Fully supervised model; their 10-shot synthetic-to-real variant reports 89.4 / 89.0, 0.33 mm and 0.90 mm.
[n] Signed along-axis tip displacement (mean ± SD 0.37 ± 1.68 mm), not an absolute 3D distance.
[l] Treatments, not patients; 7207 of 24422 needles formed the evaluation set.
[k] Strict one-to-one recall at $\tau = 50$ % (92.5 % with inner-OOF cross-fit; 94.7 % at $\tau = 35$ %); precision counts predictions matched to short obturator tracks as correct (90.0 % $F1_{adj}$, 90.5 % with inner-OOF cross-fit). Times for **NeedleFinder 2026** are full-volume uncached per-patient averages across all 54 scans on CPU (4.15 s overall: 1.26 s on the 20 standard 320 × 256 scans, 5.85 s on the 34 high-resolution 640 × 416 / 768 × 768 scans) and one RTX 4090 GPU (2.67 s overall: 0.68 s standard, 3.85 s high-resolution).

so segmentation networks reach shaft errors of 0.1 to 0.7 mm on CT [45,49,74] and 0.1 to 0.6 mm on ultrasound [56,58]. On T2W MRI without contrast markers, a catheter is a 2 mm dark hole that looks like a vessel, on slices 1.2 to 6 mm thick; the best published MRI results are 0.74 to 0.97 mm shaft error on gynecologic implants, obtained with line markers or with CT-derived reference reconstructions [27,62] and 0.75 to 0.93 mm on prostate implants [50,52]. NeedleFinder 2026 reaches a median shaft error of 0.47 mm (0.59 mm mean; 0.45 mm median / 0.57 mm mean on long active-dwell shafts) and a median 3D tip error of 2.27 mm (6.23 mm mean) on T2W MRI alone, on the largest annotated MRI cohort to date.

The head-to-head comparison in Table 2 explains why neither voxel segmentation [4,40,64] nor mask-driven tracking [74] is enough on MRI, and why the three-stage architecture works. Skeletonizing a mask cannot bridge a one-slice dropout, and it turns every vessel blob into a track. Tracking bridges the gaps but still trusts the mask. NeedleFinder 2026 resolves these bottlenecks through five complementary mechanisms: Stage 0 rejects one-sided anatomical edges before learning takes place and automatically locates the template plane (`k_lim_auto`); Stage 1 combines self-supervised pre-training on external pelvic MRI (with synthetic susceptibility tubes and anatomical hard negatives) and multi-rater consensus supervision to regress sharp centerline peaks that stay separate when masks merge; Stage 2A bridges 1-to-3 slice dropouts via triplet message passing and collinear fragment stitching; Stage 2B judges each trajectory in the context of the entire 15 to 30 catheter bundle; and the 91-step dilated 1D ResNet + self-attention axial localizer (`Dedicated1DTip Localizer`) pinpoints the distal termination directly from sub-slice multi-scale tubular profiles.

5.2 *What It Means in the Clinic, and Limitations*

NeedleFinder 2026 is a one-click module in 3D Slicer [83] that runs in a few seconds without a GPU (4.15 s mean on CPU, 1.26 s on standard scans, 2.67 s on GPU). It changes the physicist's task from tracing 15 to 30 catheters slice by slice, which takes tens of minutes, into a structured, uncertainty-guided verification step taking under a minute: click *Detect Needles*, inspect the 3D bundle, check the $k = 2$ to 8 tips with the widest split-conformal uncertainty intervals (~ 16–64 s, driving P95 tip error below 10 mm and mean tip error to 2.77 mm), and click any missing catheter when $N_{\text{pred}} < N_{\text{implant}}$ (0.65 clicks/patient, raising 1-to-1 recall to 93.0%). Because Grubbs' three-cornered-hat analysis and formal TOST tests show that the algorithm's intrinsic shaft error (0.39 mm robust SD vs. 0.35 mm for Rater 1 and 0.48 mm for Rater 2) and tip error (2.98 mm vs. 3.56 mm for Rater 2; 4.20 mm median error against independent Pelvic CT on disputed tips) are statistically non-inferior to human inter-observer variability—and 52.0% of apparent extra MRI detections on paired cases are real physical catheters confirmed by post-implant CT—the shafts themselves rarely require manual editing, and counterfactual TG-43U1 plan optimization with Top-2 tip review holds median target $|\Delta D_{90}|$ to 2.8% (5.9% mean, $\gamma_{2\%/2\text{mm}} = 74.3\%–78.2\%$).

Several limitations remain. First, while `Dedicated1DTipLocalizer` and collinear fragment stitching cut the unreviewed mean 3D tip error from 10.75 mm

(PMB 2019) and 15.65 mm (Xiao 2026) to 6.23 mm (2.27 mm median, with 58.7% of tips within ≤ 3.0 mm), 16.7% (162/972) of unreviewed matched catheters still have a tip error above 10 mm (27.27 mm P95), of which 99.3% is longitudinal stopping ambiguity along the shaft (41.4% caudal dropout undershoots, 25.3% cranial susceptibility overshoots, and 16.7% intrinsic human inter-rater tip ambiguity). Although our split-conformal uncertainty intervals (90.2% empirical coverage at 90% nominal) and $k = 5$–8 tip review workflow effectively contain this tail in practice, thin-slice isotropic 3D T2W sequences (≤ 1 mm), zero-echo-time sequences [14,17], or multi-contrast inputs are the natural next step to reduce unreviewed longitudinal tip ambiguity further. Second, while we demonstrated zero-shot generalization across 48 external `MSD Task05_Prostate` scans (0.125–0.33 FP/scan), 3T prostate brachytherapy MRI (`Case 100`, 96.3%–100.0% F1), 3D wbSSFP (92.9% F1), thick-slice 2D T2W MRI (88.4% recall), 4D serial insertion scans, and 40 paired Pelvic CT scans, all clinical brachytherapy implants come from a single institution; a prospective multi-center trial across 1.5T, 3T, and low-field systems [86] from multiple vendors is still required. Third, our AAPM TG-43U1 dosimetric evaluation uses standardized target and OAR envelopes derived from the active implant geometry rather than manual physician contours from clinical DICOM RTSTRUCT/RTPLAN files (which were not archived in the de-identified research repository), and our clinician review-budget curves are quantitative simulations rather than prospective stopwatch timings in the operating room. Fourth, our ablation shows that once multi-rater SSL Stage 1, collinear stitching, and the 1D axial tip localizer are included, `SynthBrachy3T` provides its primary detection gain in the zero-label ($k = 0$) regime (+2.85 pp F1 overall, +6.3 pp on long shafts) while still improving dwell-weighted tip error (6.65 → 5.72 mm) and counterfactual target $|\Delta D_{90}|$ (12.8% → 8.2%) at the full 43-patient training budget.

6 CONCLUSION

Evaluated under strictly nested 5 × 5 patient-level out-of-fold cross-validation with 100% reference-free axial cropping (`k_lim_auto`) across 54 clinical 3T T2W MRI scans (1,063 catheters), NeedleFinder 2026 achieves **89.1% F1** (**91.4% recall**, **86.9% precision**; **92.3% recall** and **89.1% F1** under 5 × 5 inner-OOF Stage 1 cross-fitting), 0.59 ± 0.40 **mm mean (**0.47 **mm median) in-plane shaft error**, and 6.23 ± 10.49 **mm mean (**2.27 **mm median) 3D distal tip error**, executing in 4.15 **s on an 8-core desktop CPU** (1.26 s on standard-resolution scans, 5.85 s on high-resolution scans) and 2.67 **s on a GPU**. On the 759 full-depth active-dwell shafts it reaches **91.3% recall** (94.7% at $\tau = 35\%$) and **90.0% obturator-adjusted F1** (**90.5%** with inner-OOF cross-fitting), and on the 23 patients with severe catheter adhesion it reaches **91.3%–91.8% F1**. Across all 40 doubly annotated clinical scans, Grubbs' three-cornered-hat analysis (0.39 mm intrinsic shaft SD vs. 0.35 and 0.48 mm for the two human teams), formal TOST non-inferiority tests ($p < 10^{-13}$), and independent Pelvic CT adjudication (52.0% of apparent extra MRI tracks confirmed on CT; 4.20 mm median tip error vs. 4.31 and 8.00 mm on disputed tips) establish that the algorithm is statistically non-inferior to human expert

variability. Coupled with calibrated split-conformal 3D tip uncertainty intervals (90.2% empirical coverage at 90% nominal), reviewing 2 to 8 flagged tips per patient reduces P95 tip error below 10 mm and holds counterfactual AAPM TG-43U1 ^{192}Ir target $|\Delta D_{90}|$ to 2.8% median (5.9% mean), while zero-shot transfer generalizes to 3T prostate MRI (96.3%–100.0% F1), 3D wbSSFP (92.9% F1), 4D serial insertion tracking (+5.6%–8.2% F1), and paired Pelvic CT without retraining.

DATA AVAILABILITY

The 54 clinical MRI scans and their expert traces were acquired at Brigham and Women's Hospital under Institutional Review Board approval and contain protected health information; they cannot be shared publicly. De-identified trajectory annotations and per-patient evaluation results may be made available by the corresponding author on reasonable request, subject to institutional and ethics approval. The 48 needle-free scans are the public Medical Segmentation Decathlon prostate task (Task05_Prostate), available at http://medicaldecathlon.com [81]. The SynthBrachy3T generator is released with the code, so the 300 synthetic bundles can be regenerated from a fixed random seed.

CODE AVAILABILITY

The 3D Slicer extension, the Rust Stage 0 kernel, and the exported ONNX and NumPy models are available at https://github.com/needlefinder/NeedleFinder under an open-source license. Training, nested cross-validation, statistical analysis, SynthBrachy3T, prediction-export, and figure-generation code is collected in the `vault/paper/benchmarks` directory of the same repository and will be made public on publication.

ETHICS STATEMENT

This retrospective study used imaging data acquired during routine clinical care under a protocol approved by the Institutional Review Board of Brigham and Women's Hospital / Mass General Brigham [protocol number to be inserted].

FUNDING

[Funding sources and grant numbers to be inserted.]

ACKNOWLEDGMENTS

[To be completed: clinical staff of the AMIGO suite, annotators, computing resources.]

AUTHOR CONTRIBUTIONS

[To be completed following the CRediT taxonomy.]

CONFLICTS OF INTEREST

[The authors declare no conflicts of interest, or list them here.]

REFERENCES

[1] Pötter R, Tanderup K, Kirisits C, Leeuw A de, Kirchheiner K, Nout R, Tan L T, Haie-Meder C, Mahantshetty U, Segedin B, et al 2018 The EMBRACE II study: The outcome and prospect of two decades of evolution within the GEC-ESTRO gynaecological working group and the EMBRACE studies *Clinical and Translational Radiation Oncology* **9** 48–60

[2] Pötter R, Tanderup K, Schmid M P, Jürgenliemk-Schulz I, Haie-Meder C, Fokdal L U, Sturdza A E, Hoskin P, Mahantshetty U, Segedin B, et al 2021 MRI-guided adaptive brachytherapy in locally advanced cervical cancer (EMBRACE-I): A multicentre prospective cohort study *The Lancet Oncology* **22** 538–47

[3] Mastmeyer A, Pernelle G, Ma R, Barber L and Kapur T 2017 Accurate model-based segmentation of gynecologic brachytherapy catheter collections in MRI-images *Medical Image Analysis* **42** 173–88

[4] Zaffino P, Pernelle G, Mastmeyer A, Mehrtash A, Zhang H, Kikinis R, Kapur T and Spadea M F 2019 Fully automatic catheter segmentation in MRI with 3D convolutional neural networks: Application to MRI-guided gynecologic brachytherapy *Physics in Medicine & Biology* **64** 165008

[5] Chino J, Erickson B, Gaffney D, Hsu I-C, Kamrava M, Kim Y, Niedermayr T R, Roumeliotis M, Rownd J, Todor D, Viswanathan A and Kidd E A 2025 The American Brachytherapy Society (ABS) consensus guidance for hybrid intracavitary interstitial brachytherapy for locally advanced cervical cancer *Brachytherapy* **24** 463–78

[6] JC D, P P, K T, P P, D B, C K, EM P, Limbergen E van, C H-M and R P 2012 Recommendations from gynaecological (GYN) GEC-ESTRO working group (IV): Basic principles and parameters for MR imaging within the frame of image based adaptive cervix cancer brachytherapy *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology* **103** 113–22

[7] Jacobsen M C, Beriwal S, Dyer B A, Klopp A H, Lee S I, McGinnis G J, Robbins J B, Rauch G M, Sadowski E A, Simiele S J, Stafford R J, Taunk N K, Yashar C M and Venkatesan A M 2022 Contemporary image-guided cervical cancer brachytherapy: Consensus imaging recommendations from the society of abdominal radiology and the american brachytherapy society *Brachytherapy* **21** 369–88

[8] S H, SK N, JC L, J G and K T 2009 Applicator reconstruction in MRI 3D image-based dose planning of brachytherapy for cervical cancer *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology* **91** 187–93

[9] Hellebust T P, Kirisits C, Berger D, Pérez-Calatayud J, De Brabandere M, De Leeuw A, Dumas I, Hudej R, Lowe G, Wills R and Tanderup K 2010 Recommendations from gynaecological (GYN) GEC-ESTRO working group: Considerations and pitfalls in commissioning and applicator reconstruction in 3D image-based treatment planning of cervix cancer brachytherapy *Radiotherapy and Oncology* **96** 153–60

[10] Ladbury C, Harkenrider M, Taunk N, Fisher C, Mayadev J, Venkat P, Yashar C, Gaffney D, Beriwal S and Glaser S 2023 A practical guide to hybrid interstitial/intracavitary brachytherapy for locally-advanced cervical cancer *Brachytherapy* **22** 640–8

[11] Prisciandaro J, Zoberi J E, Cohen G, Kim Y, Johnson P, Paulson E, Song W, Hwang K-P, Erickson B, Beriwal S, Kirisits C and Mourtada F 2022 AAPM task group report 303 endorsed by the ABS: MRI implementation in HDR brachytherapy—considerations from simulation to treatment *Medical Physics* **49** e983–1023

[12] Richardson S L, Baghwala A, Cohen G N, Dempsey C, Libby B, Melhus C S, Miller R A, Scanderbeg D J and Simiele S J 2025 AAPM medical physics practice guideline 13.a: HDR brachytherapy, part B *Journal of Applied Clinical Medical Physics* **26** e70118

[13] Shaaer A, Paudel M, Smith M, Tonolete F, Nicolae A, Leung E and Ravi A 2020 Evaluation of an MR-only interstitial gynecologic brachytherapy workflow using MR-line marker for catheter reconstruction *Brachytherapy* **19** 642–50

[14] Lee C Y, Kaza E, Harris T C, O'Farrell D A, King M T, Dyer M A, Cormack R A and Buzurovic I 2023 Catheter reconstruction and dosimetric verification of MRI-only treatment planning (MRTP) for interstitial HDR brachytherapy using PETRA sequence *Physics in Medicine & Biology* **68** 035002

[15] Ning M S, Vedam S, Ma J, Stafford R J, Bruno T L, Cunningham M, Starks C, Lawyer A, Venkatesan A M, Wang L, Wang J, Olivieri N D, Guzman A B, Incalcaterra J R, Thaker N G, Joyner M M, Lin L L, Jhingran A, Eifel P J and Klopp A H 2020 Clinical utility and value contribution of an MRI-positive line marker for image-guided brachytherapy in gynecologic malignancies *Brachytherapy* **19** 305–15

[16] Zhang H, Donnelly E D, Strauss J B, Kang Z, Gopalakrishnan M, Lee P C, Khelashvili G, Nair C K, Lee B H and Sathiaseelan V 2019 Clinical implementation, logistics and workflow guide for MRI image based interstitial HDR brachytherapy for gynecological cancers *Journal of applied clinical medical physics* **20** 37–49

[17] Kaza E, Lee C Y, King M T, Dyer M A, Cormack R A and Buzurovic I 2022 First pointwise encoding time reduction with radial acquisition (PETRA) implementation for catheter detection in interstitial high-dose-rate (HDR) brachytherapy *Brachytherapy* **21** 501–10

[18] Richart J, Carmona-Meseguer V, García-Martínez T, Herreros A, Otal A, Pellejero S, Tornero-López A and Pérez-Calatayud J 2018 Review of strategies for MRI based reconstruction of endocavitary and interstitial applicators in brachytherapy of cervical cancer *Reports of practical oncology and radiotherapy : journal of Greatpoland Cancer Center in Poznan and Polish Society of Radiation Oncology* **23** 547–61

[19] Malajovich I, Anamalayil S, Dolney O V, Kevin Teo B K, Arscott W T and Taunk N K 2020 Techniques for and uncertainties of MRI-based reconstruction of titanium tandem and ring brachytherapy applicators *Brachytherapy* **19** 651–8

[20] Sargazi V, Naseri S, Gholamhosseinian H and Momennezhad M 2026 Applicator reconstruction in cervical cancer brachytherapy: A systematic review of current methods, challenges, and AI-driven future directions *Brachytherapy* **25** 262–74

[21] Tanderup K, Hellebust T P, Lang S, Granfeldt J, Pötter R, Lindegaard J C and Kirisits C 2008 Consequences of random and systematic reconstruction uncertainties in 3D image based brachytherapy in cervical cancer *Radiotherapy and Oncology* **89** 156–63

[22] Schindel J, Zhang W, Bhatia S K, Sun W and Kim Y 2013 Dosimetric impacts of applicator displacements and applicator reconstruction-uncertainties on 3D image-guided brachytherapy for cervical cancer *Journal of Contemporary Brachytherapy* **5** 250–7

[23] Singh A, Karshafian R, Ravi A and Paudel M R 2026 Dosimetric evaluation of systematic source position uncertainty along the source path and robust planning during MRI-only cervical HDR brachytherapy *Journal of Applied Clinical Medical Physics* **27** e70810

[24] Nunez D A, Trager M, Beaudry J, Cohen G N, Dauer L T, Gorovets D, Hassan Rezaeian N, Kollmeier M A, Leong B, McCann P, Williamson M, Zelefsky M J and Damato A L 2021 Quantifying clinical severity of physics errors in high-dose rate prostate brachytherapy using simulations *Brachytherapy* **20** 1062–9

[25] Rylander S, Buus S, Pedersen E M, Bentzen L and Tanderup K 2017 Dosimetric impact of contouring and needle reconstruction uncertainties in US-, CT- and MRI-based high-dose-rate prostate brachytherapy treatment planning *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology* **123** 125–32

[26] Androulakis I, Godart J, Luthart L, MEMC C, Westerveld H, Nout R A, Hoogeman M and Kolkman-Deurloo I K 2025 Reconstruction errors in clinical intraoperative TRUS-based prostate HDR-BT detected using electromagnetic tracking *Brachytherapy* **24** 177–85

[27] Shaaer A, Paudel M, Smith M, Tonolete F and Ravi A 2022 Deep-learning-assisted algorithm for catheter reconstruction during MR-only gynecological interstitial brachytherapy *Journal of Applied Clinical Medical Physics* **23** e13494

[28] Roumeliotis M, Antaki M, Kim G, Gardner U G, Sheikh K, Lee J, Seethamraju R, Tokuda J, Hu C, Schmidt E and Viswanathan A N 2026 Prospective operational feasibility, safety, and workflow of magnetic resonance-guided tracking technology for interstitial gynecologic brachytherapy *Brachytherapy* **25** 741–8

[29] Vijayan S, Theeler J J, Fu Z and Kim Y 2026 Systematic review of artificial intelligence in brachytherapy *Journal of Contemporary Brachytherapy* **18** 85–114

[30] Kapur T, Egger J, Damato A, Schmidt E J and Viswanathan A N 2012 3-t MR-guided brachytherapy for gynecologic malignancies *Magnetic resonance imaging* **30** 1279–90

[31] Chen X and Egger J 2014 Development of an open source software module for enhanced visualization during MR-guided interstitial gynecologic brachytherapy *SpringerPlus* **3** 167

[32] Pernelle G, Mehrtash A, Barber L, Damato A, Wang W, Seethamraju R T, Schmidt E J, Cormack R A, Wells W M, Viswanathan A N and Kapur T 2013 Validation of catheter segmentation for MR-guided gynecologic cancer brachytherapy *Medical image computing and computer-assisted intervention – MICCAI 2013* Lecture notes in computer science vol 8151 (Springer) pp 380–7

[33] Mastmeyer A, Pernelle G, Barber L, Pieper S, Fortmeier D, Wells S, Handels H and Kapur T 2017 Model-based catheter segmentation in MRI-images

[34] J R, A O, S R, AI N, M D, M S, J V, F B and J P-C 2015 A practical MRI-based reconstruction method for a new endocavitary and interstitial gynaecological template *Journal of contemporary brachytherapy* **7** 407–14

[35] W Q, M Y, M D, D T and A F 2013 Needle segmentation using 3D hough transform in 3D TRUS guided prostate transperineal therapy *Medical physics* **40** 042902

[36] Dise J, Liang X, Scheuermann J, Anamalayil S, Mesina C, Lin L L and Teo B K 2015 Development and evaluation of an automatic interstitial catheter digitization tool for adaptive high-dose-rate brachytherapy *Brachytherapy* **14** 619–25

[37] Deufel C L, Tian S, Yan B B, Vaishnav B D, Haddock M G and Petersen I A 2020 Automated applicator digitization for high-dose-rate cervix brachytherapy using image thresholding and density-based clustering *Brachytherapy* **19** 111–8

[38] Hrinivich W T, Morcos M, Viswanathan A and Lee J 2019 Automatic tandem and ring reconstruction using MRI for cervical cancer brachytherapy *Medical Physics* **46** 4324–32

[39] Zheng Y, Jiang S, Yang Z and Wei L 2021 Automatic needle detection using improved random sample consensus in CT image-guided lung interstitial brachytherapy *Journal of Applied Clinical Medical Physics* **22** 121–31

[40] Isensee F, Jaeger P F, Kohl S A A, Petersen J and Maier-Hein K H 2021 nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation *Nature Methods* **18** 203–11

[41] Jung H, Shen C, Gonzalez Y, Albuquerque K and Jia X 2019 Deep-learning assisted automatic digitization of interstitial needles in 3D CT image based high dose-rate brachytherapy of gynecological cancer *Physics in Medicine & Biology* **64** 215003

[42] Jung H, Gonzalez Y, Shen C, Klages P, Albuquerque K and Jia X 2019 Deep-learning-assisted automatic digitization of applicators in 3D CT image-based high-dose-rate brachytherapy of gynecological cancer *Brachytherapy* **18** 841–51

[43] Zhang D, Yang Z, Jiang S, Zhou Z, Meng M and Wang W 2020 Automatic segmentation and applicator reconstruction for CT-based brachytherapy of cervical cancer using 3D convolutional neural networks *Journal of Applied Clinical Medical Physics* **21** 158–69

[44] Hu H, Yang Q, Li J, Wang P, Tang B, Wang X and Lang J 2021 Deep learning applications in automatic segmentation and reconstruction in CT-based cervix brachytherapy *Journal of Contemporary Brachytherapy* **13** 325–30

[45] Weishaupt L L, Sayed H K, Mao X, Choo R, Stish B J, Enger S A and Deufel C L 2022 Approaching automated applicator digitization from a new angle: Using sagittal images to improve deep learning accuracy and robustness in high-dose-rate prostate brachytherapy *Brachytherapy* **21** 520–31

[46] Xie H, Wang J, Chen Y, Tu Y, Chen Y, Zhao Y, Zhou P, Wang S, Bai Z and Tang Q 2023 Automatic reconstruction of interstitial needles using CT images in post-operative cervical cancer brachytherapy based on deep learning *Journal of Contemporary Brachytherapy* **15** 134–40

[47] Wang Y, Jian W, Zhu L, Cai C, Zhang B and Wang X 2024 Attention-gated deep-learning-based automatic digitization of interstitial needles in high-dose-rate brachytherapy for cervical cancer *Advances in Radiation Oncology* **9** 101340

[48] Moradi M M, Siavashpour Z, Takhtardeshir S, Showkatian E, Jaberi R, Ghaderi R, Mofid B and Taghizadeh-Hesary F 2025 Fully automatic reconstruction of prostate high-dose-rate brachytherapy interstitial needles using two-phase deep learning-based segmentation and object tracking algorithms *Clinical and Translational Radiation Oncology* **51** 100925

[49] Quetin S, Jafarzadeh H, Kalinowski J, Bekerat H, Bahoric B, Maleki F and Enger S A 2025 Automatic catheter digitization in breast brachytherapy *Medical Physics* **52** e18107

[50] Dai X, Lei Y, Zhang Y, Qiu R L J, Wang T, Dresser S A, Curran W J, Patel P, Liu T and Yang X 2020 Automatic multi-catheter detection using deeply supervised convolutional neural network in MRI-guided high-dose-rate prostate brachytherapy *Medical Physics* **47** 4115–24

[51] Aleong A M, Berlin A, Borg J, Helou J, Beiki-Ardakani A, Rink A, Raman S, Chung P and Weersink R A 2024 Rapid multi-catheter segmentation for magnetic resonance image-guided catheter-based interventions *Medical Physics* **51** 5361–73

[52] Kostoulas V, Guijt A, Kerkhof E M, Pieters B R, Bosman P A N and Alderliesten T 2025 Dealing with segmentation errors in needle reconstruction for MRI-guided brachytherapy *Medical imaging 2025: Image-guided procedures, robotic interventions, and modeling, proc. SPIE 13408* p 1340826

[53] Li X, Young A S, Raman S S, Lu D S, Lee Y H, Tsao T C and Wu H H 2020 Automatic needle tracking using mask r-CNN for MRI-guided percutaneous interventions *International journal of computer assisted radiology and surgery* **15** 1673–84

[54] Zhou W, Li X, Zabihollahy F, Lu D S and Wu H H 2024 Deep learning-based automatic pipeline for 3D needle localization on intra-procedural 3D MRI *International Journal of Computer Assisted Radiology and Surgery* **19** 2227–37

[55] Zhang Y, He X, Tian Z, Jeong J J, Lei Y, Wang T, Zeng Q, Jani A B, Curran W J, Patel P, Liu T and Yang X 2020 Multi-needle detection in 3D ultrasound images using unsupervised order-graph regularized sparse dictionary learning *IEEE transactions on medical imaging* **39** 2302–15

[56] Zhang Y, Tian Z, Lei Y, Wang T, Patel P, Jani A B, Curran W J, Liu T and Yang X 2020 Automatic multi-needle localization in ultrasound images using large margin mask RCNN for ultrasound-guided prostate brachytherapy *Physics in Medicine & Biology* **65** 205003

[57] Zhang Y, Lei Y, Qiu R L J, Wang T, Wang H, Jani A B, Curran W J, Patel P, Liu T and Yang X 2020 Multi-needle localization with attention u-net in US-guided HDR prostate brachytherapy *Medical physics* **47** 2735–45

[58] Andersén C, Rydén T, Thunberg P and Lagerlöf J H 2020 Deep learning-based digitization of prostate brachytherapy needles in ultrasound images *Medical Physics* **47** 6414–20

[59] Wang F, Xing L, Bagshaw H, Buyyounouski M and Han B 2020 Deep learning applications in automatic needle segmentation in ultrasound-guided prostate brachytherapy *Medical Physics* **47** 3797–805

[60] Liu D, Tupor S, Singh J, Chernoff T, Leong N, Sadikov E, Amjad A and Zilles S 2022 The challenges facing deep learning-based catheter localization for ultrasound guided high-dose-rate prostate brachytherapy *Medical Physics* **49** 2442–51

[61] Goulet M, Duguay-Drouin P, Mascolo-Fortin J, Mégrourèche J, Octave N and Tsui J M G 2025 Clinical application of deep learning-assisted needles reconstruction in prostate ultrasound brachytherapy *International Journal of Radiation Oncology, Biology, Physics* **122** 199–207

[62] Kim G, Antaki M, Schmidt E J, Roumeliotis M, Viswanathan A N and Lee J 2026 Automatic digitization of applicator and catheters for MRI-guided cervical cancer brachytherapy *Brachytherapy* **25** 85–94

[63] Hu X, Li F, Samaras D and Chen C 2019 Topology-preserving deep image segmentation *Advances in neural information processing systems 32 (NeurIPS 2019)* vol 32

[64] Shit S, Paetzold J C, Sekuboyina A, Ezhov I, Unger A, Zhylka A, Pluim J P W, Bauer U and Menze B H 2021 clDice – a novel topology-preserving loss function for tubular structure segmentation *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* pp 16555–64

[65] Kirchhoff Y, Rokuss M R, Roy S, Kovacs B, Ulrich C, Wald T, Zenk M, Vollmuth P, Kleesiek J, Isensee F and Maier-Hein K 2024 Skeleton recall loss for connectivity conserving and resource efficient segmentation of thin tubular structures *Computer vision – ECCV 2024, lecture notes in computer science* (Springer) pp 218–34

[66] Zhou W, Dai Q, Curiel O, Tsao T C, Chiang J, Lu D S and Wu H H 2026 Keypoint detection network for needle localization on intra-procedural MRI in MRI-guided liver interventions *International journal of computer assisted radiology and surgery* **21** 1775–88

[67] Mehrtash A, Damato A, Pernelle G, Barber L, Farhat N, Viswanathan A, Cormack R and Kapur T 2014 EM-navigated catheter placement for gynecologic brachytherapy: An accuracy study *Medical imaging 2014: Image-guided procedures, robotic interventions, and modeling, proc. SPIE 9036* vol 9036 p 90361F

[68] Wang W, Viswanathan A N, Damato A L, Chen Y, Tse Z, Pan L, Tokuda J, Seethamraju R T, Dumoulin C L, Schmidt E J and Cormack R A 2015 Evaluation of an active magnetic resonance tracking system for interstitial brachytherapy *Medical physics* **42** 7114–21

[69] Arcos J de, Schmidt E J, Wang W, Tokuda J, Vij K, Seethamraju R T, Damato A L, Dumoulin C L, Cormack R A and Viswanathan A N 2017 Prospective clinical implementation of a novel magnetic resonance tracking device for real-time brachytherapy catheter positioning *International journal of radiation oncology, biology, physics* **99** 618–26

[70] Dürrbeck C, Sauer B C, Lotter M, Kreppner S, Strnad V, Fietkau R and Bert C 2022 Automating implant reconstruction in interstitial brachytherapy of the breast: A hybrid approach combining electromagnetic tracking and image segmentation *Radiotherapy and Oncology* **176** 172–8

[71] Cantin A, Lavallée M-C, Poulin E, Hautvast G, Foster W and Beaulieu L 2024 Proof-of-concept of real-time electromagnetic guidance for gynecologic interstitial catheters in high dose rate brachytherapy *Physics and Imaging in Radiation Oncology* **32** 100661

[72] Deufel C L, Brost E E, Dupere J M, Petersen I A, Haddock M G and Garda A E 2024 EMVision: An electromagnetic tracking guidance system for accurate placement of interstitial brachytherapy applicators *Brachytherapy* **23** 676–86

[73] Beaulieu L, Bert C, Borot de Battisti M, Cormack R A, Cunha J A M, Damato A L, Deufel C L, Hautvast G L T F, Hsu I-C, Kolkman-Deurloo I-K, Moerland M A, Niatsetski Y, Weersink R A and Tanderup K 2025 AAPM task group 317 report: A joint AAPM and ESTRO report on brachytherapy catheter, needle, and applicator tracking technology *Medical Physics* **52** e70037

[74] Xiao Z, Liu B, Wang J, Yao Q, Sun H, Zhou F, Wang J, Wu Q and Jiang P 2026 Multi-catheter digitization in brachytherapy via few-shot synthetic-to-real learning and structure-aware tracking *arXiv preprint arXiv:2607.11290*

[75] Xiao Z, Zhou F, Wang J, He C, Liu B, Sun H, Ji Z, Jiang Y, Wang J and Wu Q 2026 Multi-needle localization for pelvic seed implant brachytherapy based on tip-handle detection and matching *IEEE Journal of Biomedical and Health Informatics* 1–14

[76] Brasó G and Leal-Taixé L 2020 Learning a neural solver for multiple object tracking *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* pp 6246–56

[77] Ben-Haim T and Riklin Raviv T 2022 Graph neural network for cell tracking in microscopy videos *Computer vision – ECCV 2022* Lecture notes in computer science vol 13681 (Springer) pp 610–26

[78] Shit S, Koner R, Wittmann B, Paetzold J, Ezhov I, Li H, Pan J, Sharifzadeh S, Kaissis G, Tresp V and Menze B 2022 Relationformer: A unified framework for image-to-graph generation *Computer vision – ECCV 2022, lecture notes in computer science* (Springer) pp 422–39

[79] Arapi V, Hardt-Stremayr A, Weiss S and Steinbrener J 2023 Bridging the simulation-to-real gap for AI-based needle and target detection in robot-assisted ultrasound-guided interventions *European Radiology Experimental* **7** 30

[80] Fechter T, Sachpazidis I and Baltas D 2024 The use of deep learning in interventional radiotherapy (brachytherapy): A review with a focus on open source and open data *Zeitschrift für Medizinische Physik* **34** 180–96

[81] Antonelli M, Reinke A, Bakas S, Farahani K, Kopp-Schneider A, Landman B A, Litjens G, Menze B, Ronneberger O, Summers R M, et al 2022 The medical segmentation decathlon *Nature Communications* **13** 4128

[82] C K, MJ R, D B, F B, M D B, Laarse R van der, Y N, P P, TP H, J P-C, K T, JL V and FA S 2014 Review of clinical brachytherapy uncertainties: Analysis guidelines of GEC-ESTRO and the AAPM *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology* **110** 199–212

[83] Fedorov A, Beichel R, Kalpathy-Cramer J, Finet J, Fillion-Robin J-C, Pujol S, Bauer C, Jennings D, Fennessy F, Sonka M, et al 2012 3D Slicer as an image computing platform for the quantitative imaging network *Magnetic Resonance Imaging* **30** 1323–41

[84] Kallis K, Kaltsas T, Kreppner S, Lotter M, Strnad V, Fietkau R and Bert C 2019 Impact of inter- and intra-observer variabilities of catheter reconstruction on multi-catheter interstitial brachytherapy of breast cancer patients *Radiotherapy and Oncology* **135** 25–32

[85] Karius A, Kreppner S, Strnad V, Schweizer C, Lotter M, Fietkau R and Bert C 2024 Inter-observer effects in needle reconstruction for temporary prostate brachytherapy: Dosimetric implications and adaptive CBCT-TRUS registration solutions *Brachytherapy* **23** 421–32

[86] Major T, Vízkeleti J and Takácsi-Nagy Z 2026 Geometric accuracy of applicator and needle reconstruction in gynecological brachytherapy using low-field magnetic resonance imaging *Journal of Contemporary Brachytherapy* **18** 288–95